FACULTY OF COMPUTING
UTM Johor Bahru

SECD2613: System Analysis and Design

2025/2026-2

ANALYSIS AND DESIGN
FlexHub AI Marketplace (FlexServe)

Phase 3

Section : 01
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1. Overview of the Project

Service providers ranging from makeup artists and consultants to private tutors are rapidly expanding. However, the digital platforms for supporting this expertise often rely on rigid, traditional booking models. Existing platforms fail to help service providers to better manage their time and connect users to the services they need. FlexHub (FlexServe) aims to solve this issue by implementing an AI time management system and better help users match a service provider based on their time and need. This platform will facilitate seamless connections between clients searching for specific expertise and freelancers offering their services by the hour or by the session.

2. Problem Statement

Most commonly, clients seeking professional services must rely on a fragmented ecosystem, as each service provider has their own isolated social media pages or static booking websites. The lack of a centralized website hosting huge ranges of different services makes it difficult for a client to discover experts who match their specific needs and schedule constraints. For example, if someone wants to hire a landscaper to help decorate their garden and rely on the few social media pages they follow, they will have very limited choices to choose from causing they might be unable to find the most suitable landscaper or have a hard time scheduling appointments with the landscaper they might choose.

The few centralized websites or platforms that exist are, however, too static. Consequently, both professional experts and the client using the platforms struggle with inefficient scheduling systems. Static calendars frequently lead to double bookings, and also unexpected scheduling conflicts that happen because their system is unable to factor in buffer times between appointments. These legacy systems fail to account for the required buffer times, such as considering the travel between client locations, or a human requirement of the preparation and cleanup between sessions. Furthermore, existing platforms strictly categorize users as either service providers or clients only, so there is no flexibility for individuals who wish to operate as service providers and become consumers for other type of services officered within the same ecosystem.

3. Proposed Solutions

We propose developing FlexServe, a sub-system of the overarching FlexHub AI Marketplace, designed specifically to manage professional experts acting as freelancers. This centralized platform will empower users to seamlessly discover, match, and book experts based on availability and preferences.

To address the limitations of static platforms, the system will incorporate one core AI integrations:

- **AI Smart Scheduler:** To eliminate booking conflicts, the system will utilize an intelligent time-slot management algorithm. This AI feature will dynamically calculate and enforce "buffer times" between bookings, automatically accounting for estimated travel time or session cleanup, ensuring a realistic and stress-free schedule for the provider.

Additionally, the platform will analyze user preferences, past interactions, and service requirements to recommend the most suitable experts, ensuring a high-quality match between the client's needs and the provider's skills. The system will also feature Role Management, allowing a single user account to seamlessly toggle between being a Client and Service Provider.

4. Current Business Process/Workflow

4.1 Current Business Process

Based on AS-IS analysis

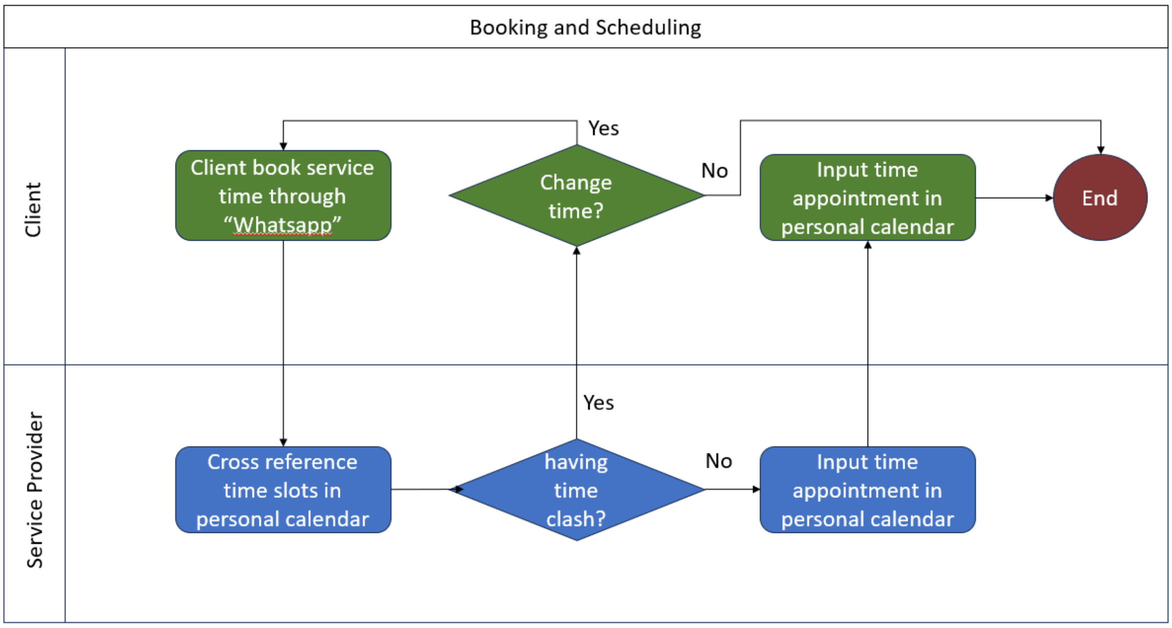

Figure 4.1.1 Booking and scheduling flowchart

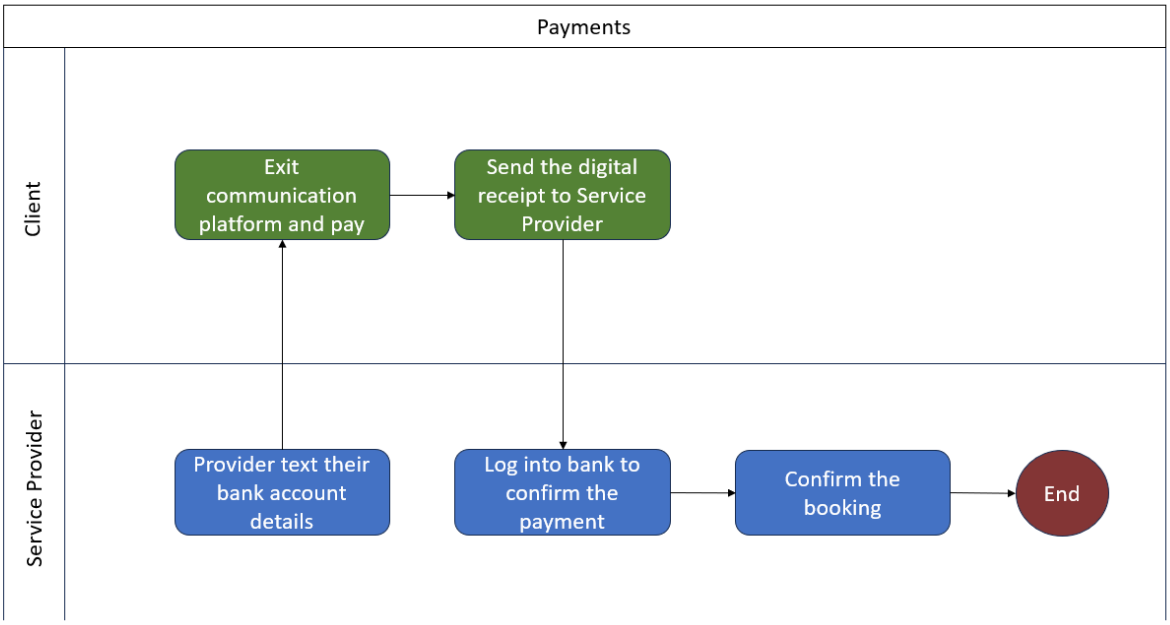

Figure 4.1.2 Payments flowchart

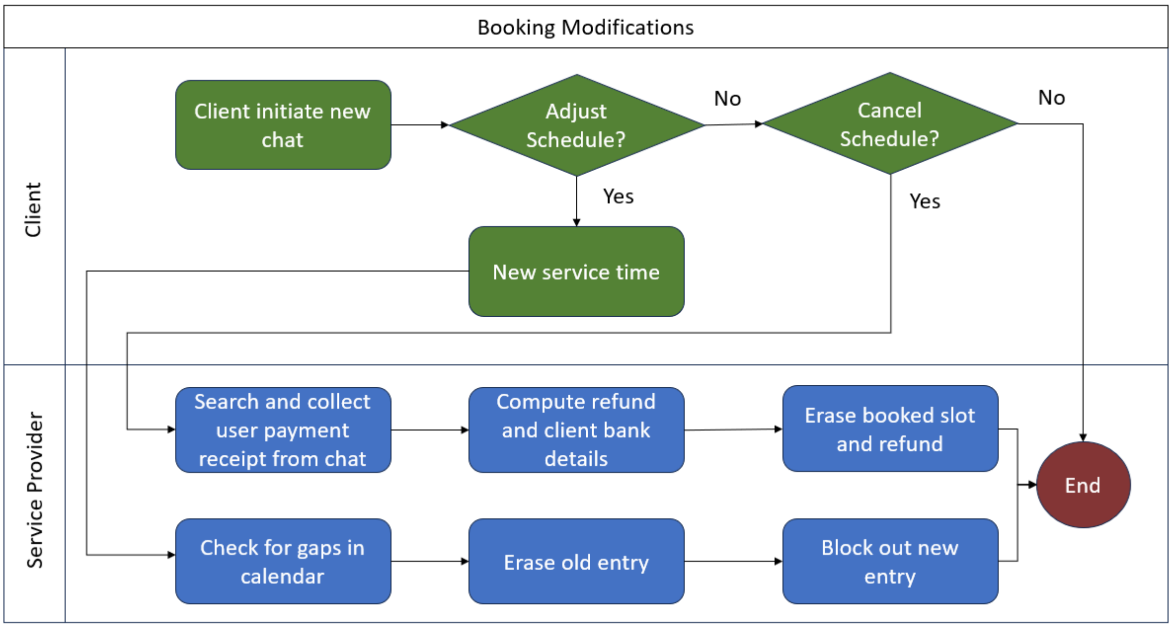

Figure 4.1.3 Booking modifications, cancellation and refunds flowchart

5. Logical DFD (AS-IS)

5.1 Logical DFD AS-IS system

Context Diagram:

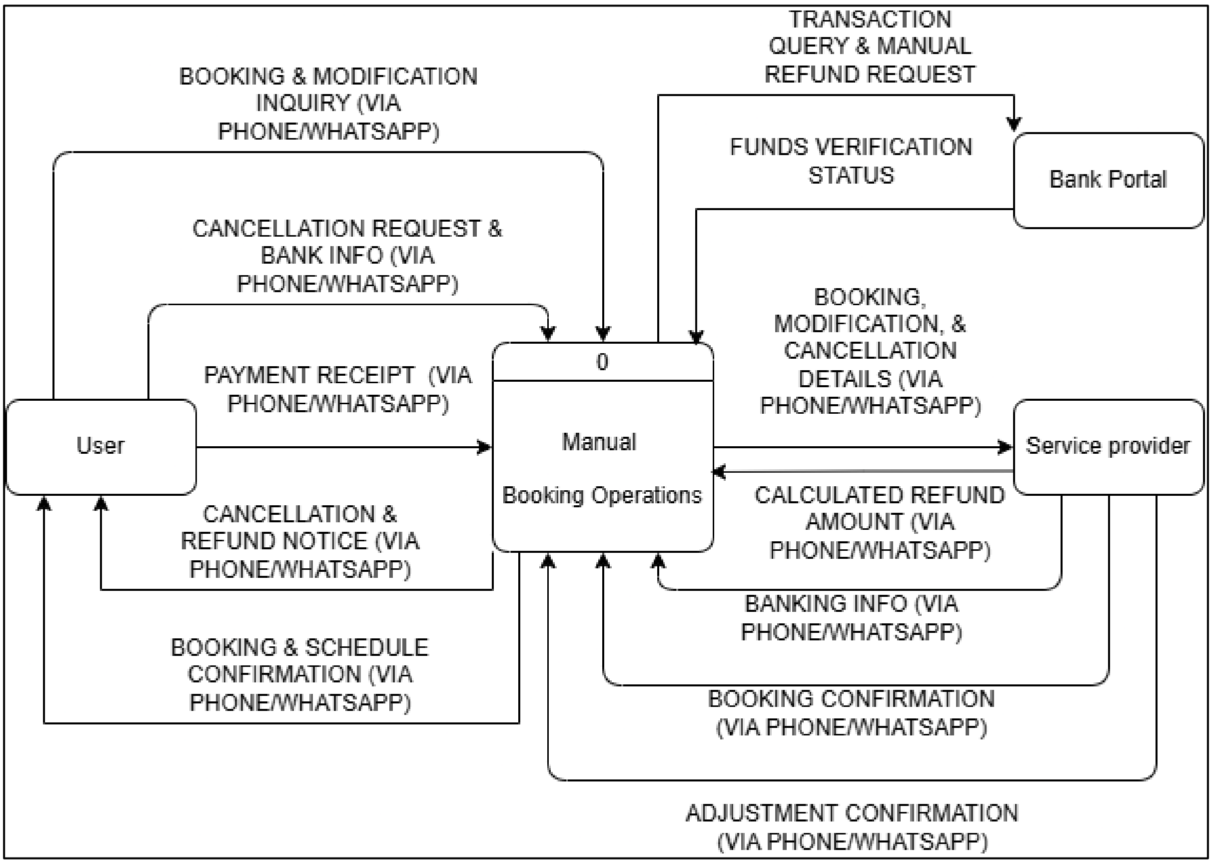

Figure 5.1.1 Context diagram of existing system

Diagram 0:

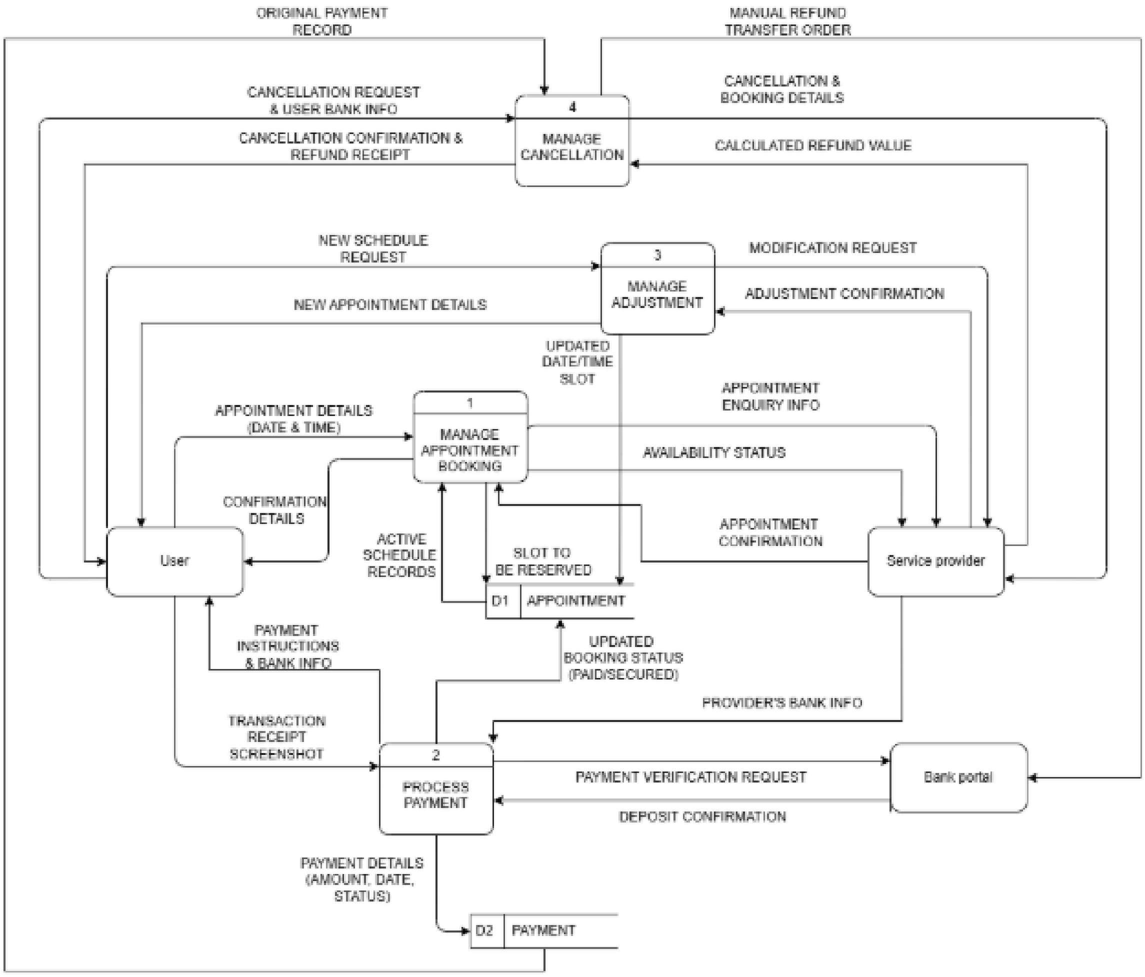

Figure 5.1.2 Diagram 0 of existing system

Child diagram:

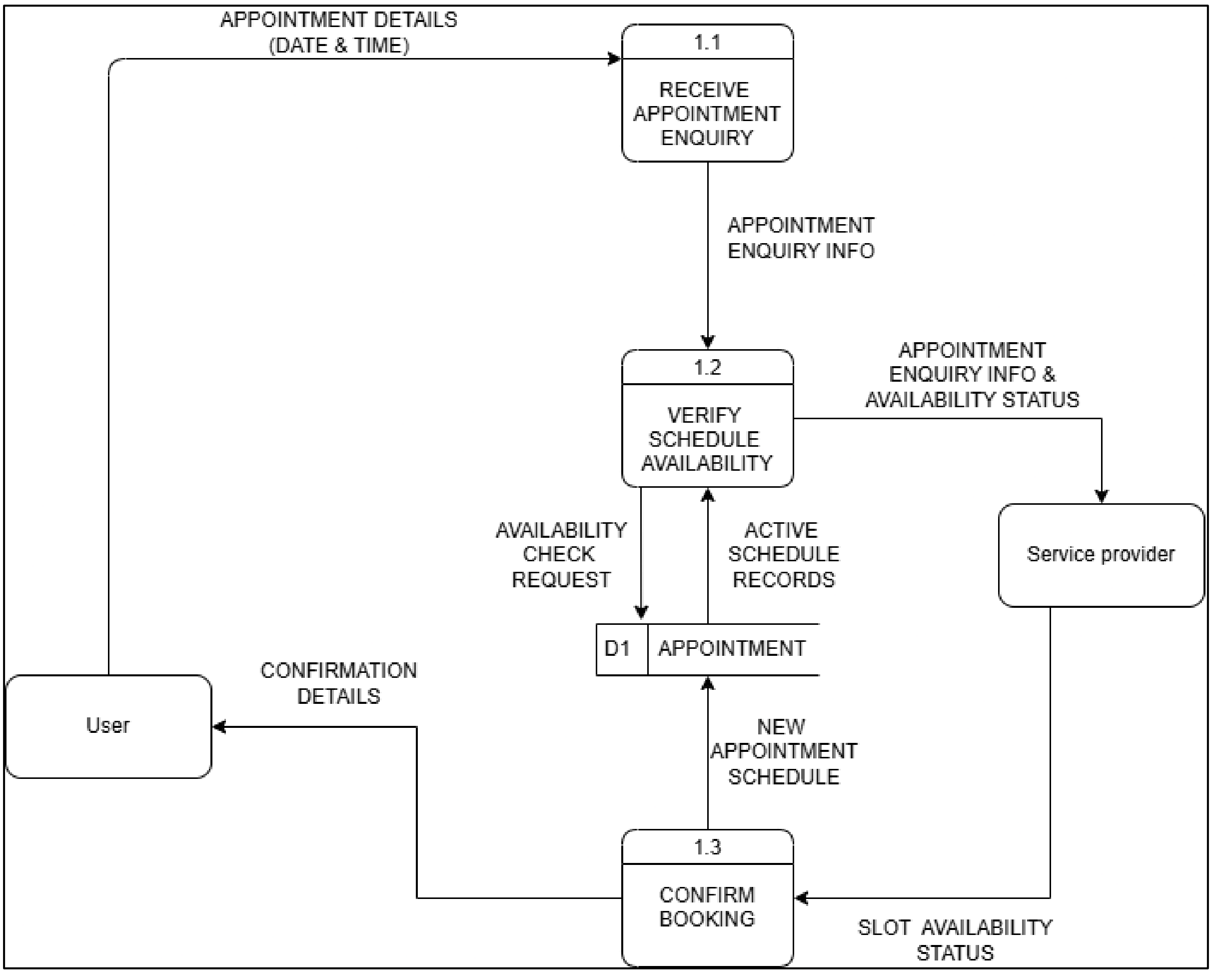

Figure 5.1.3 Child diagram of Process 1

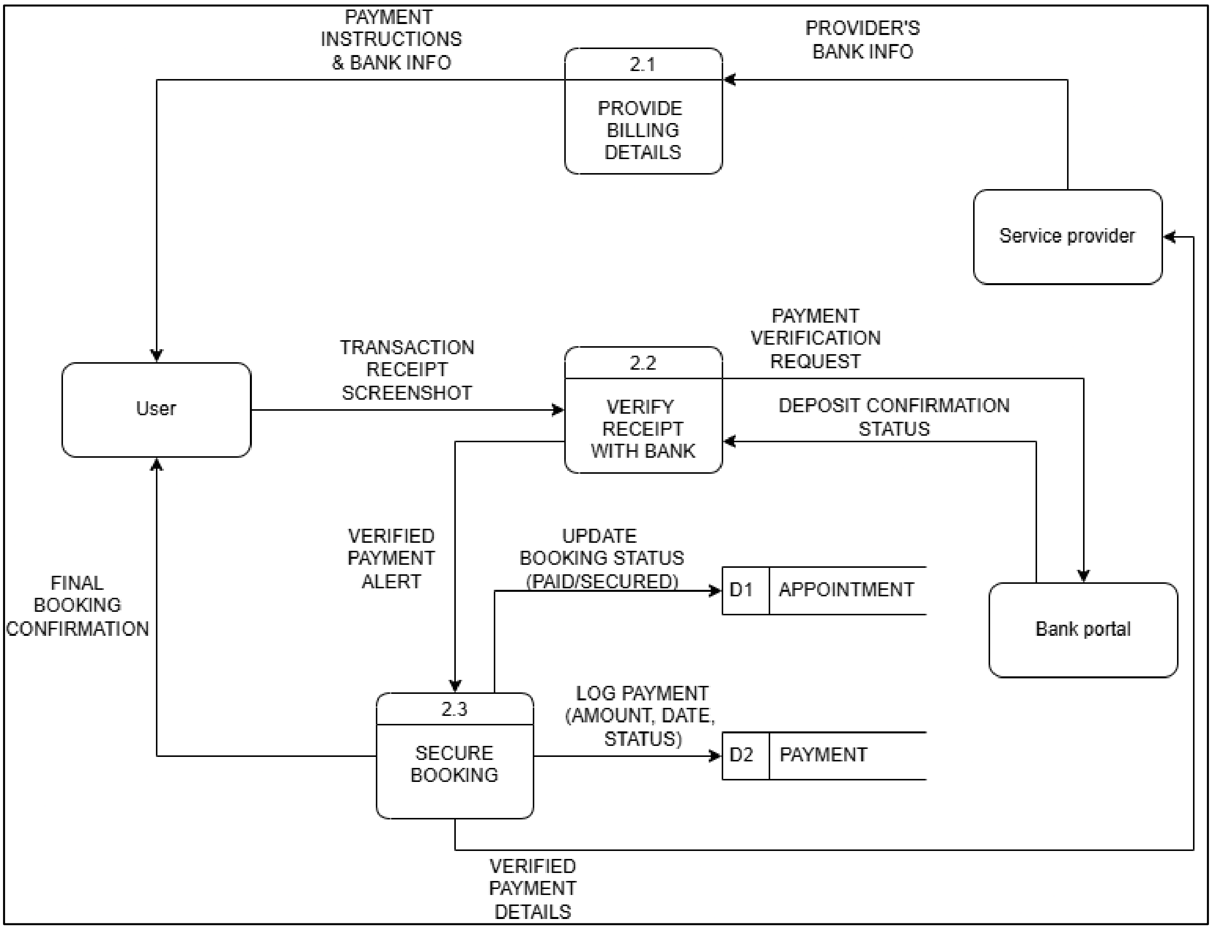

Figure 5.1.4 Child diagram of Process 2

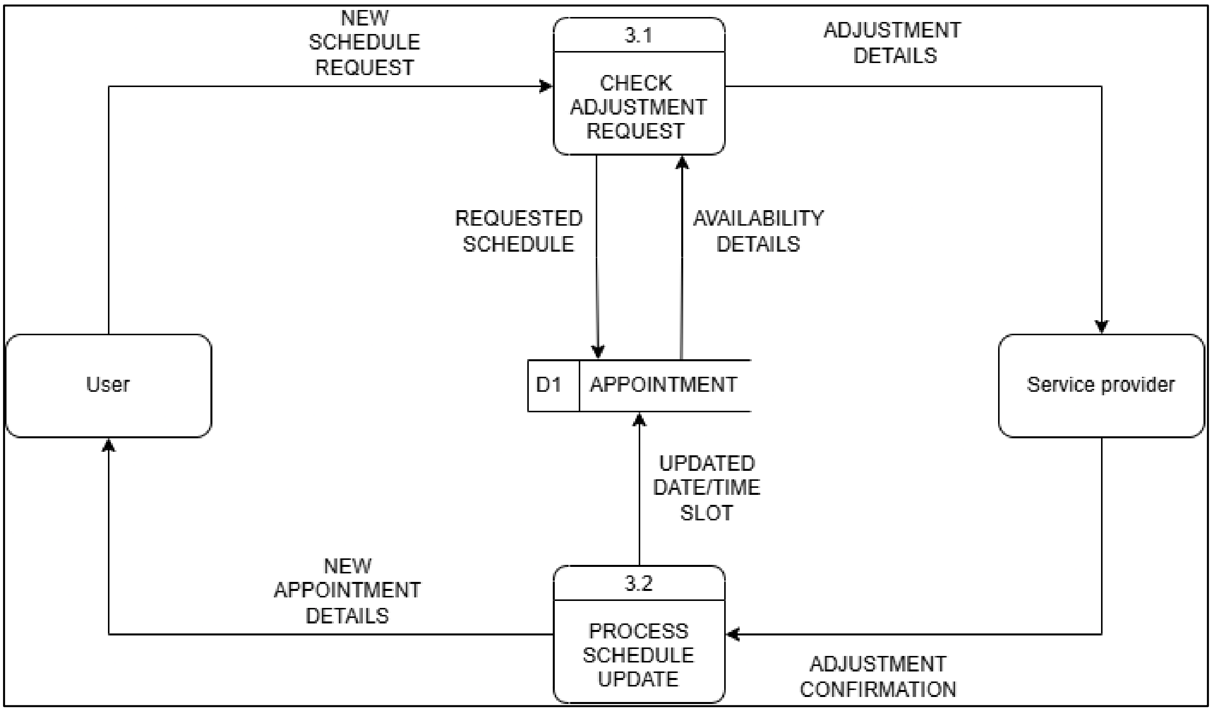

Figure 5.1.5 Child diagram of Process 3

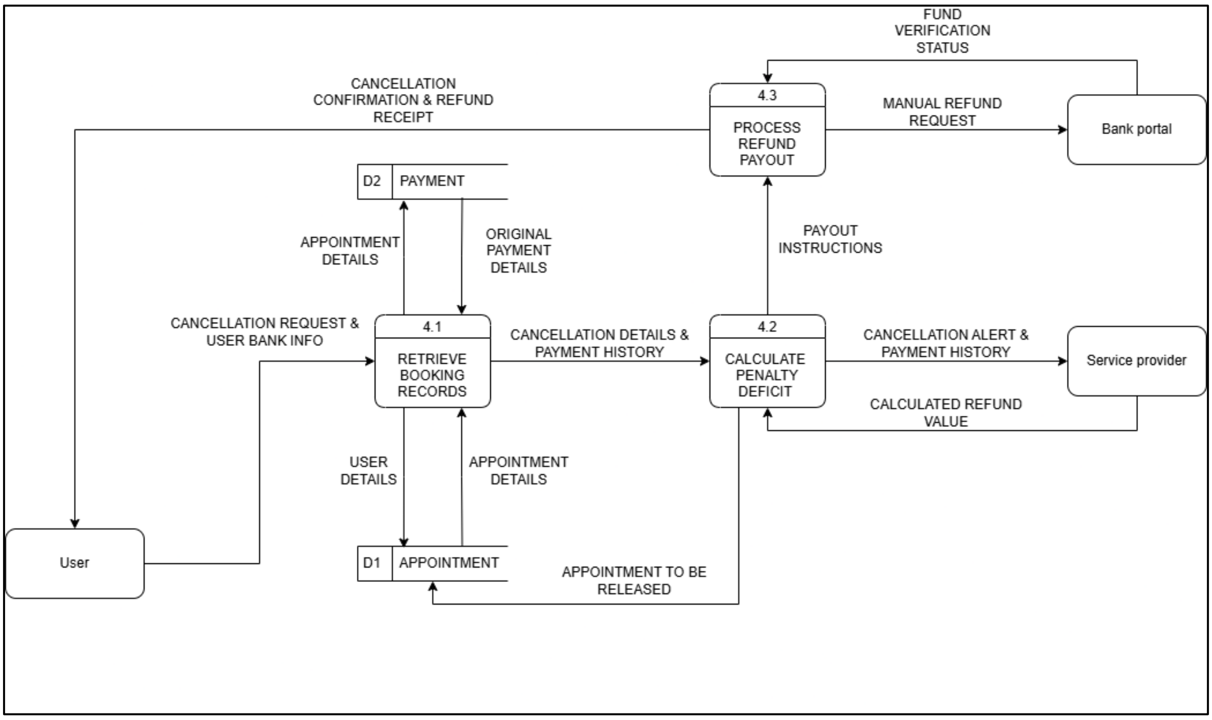

Figure 5.1.6 Child diagram of Process 4

6. System Analysis and Specification

6.1 Logical DFD TO-BE system

Context Diagram:

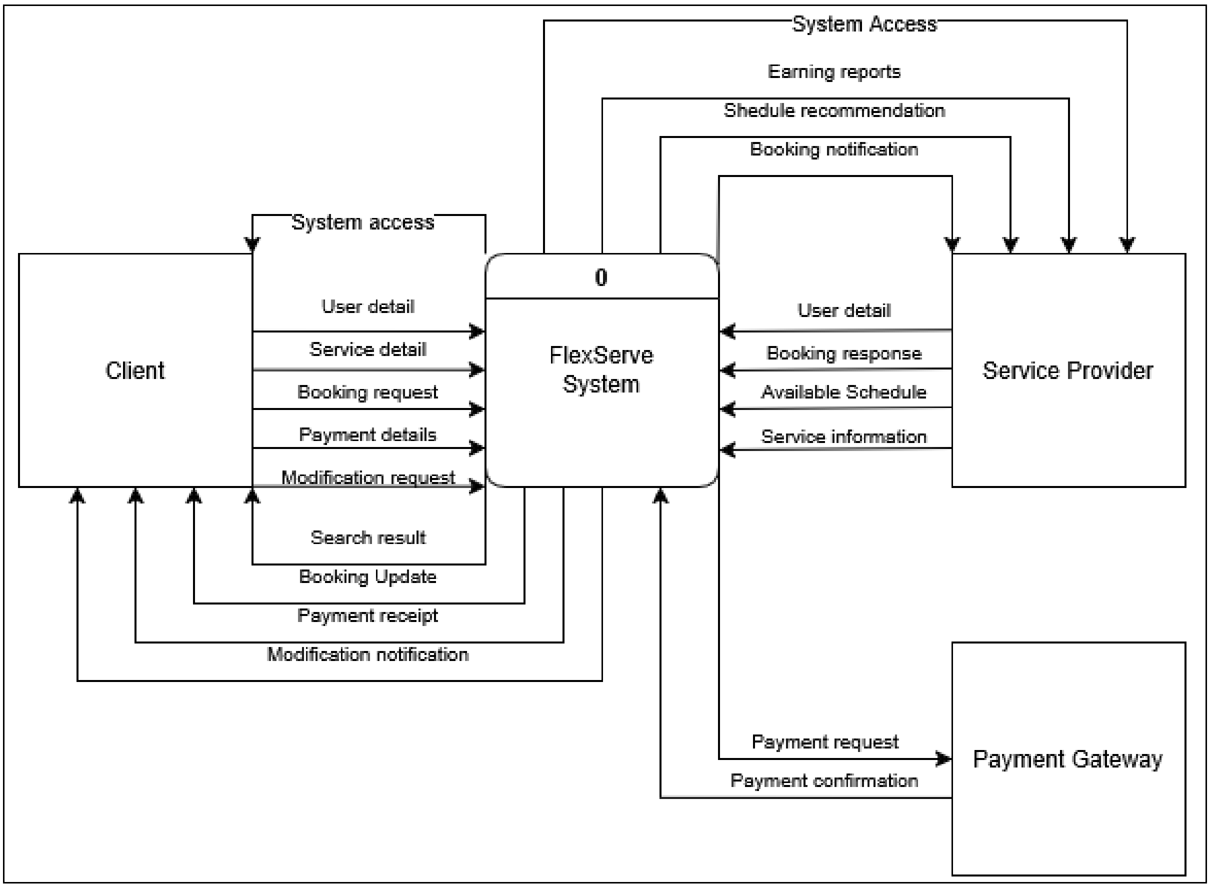

Figure 6.1.1 Context diagram of TO-BE system

Diagram 0:

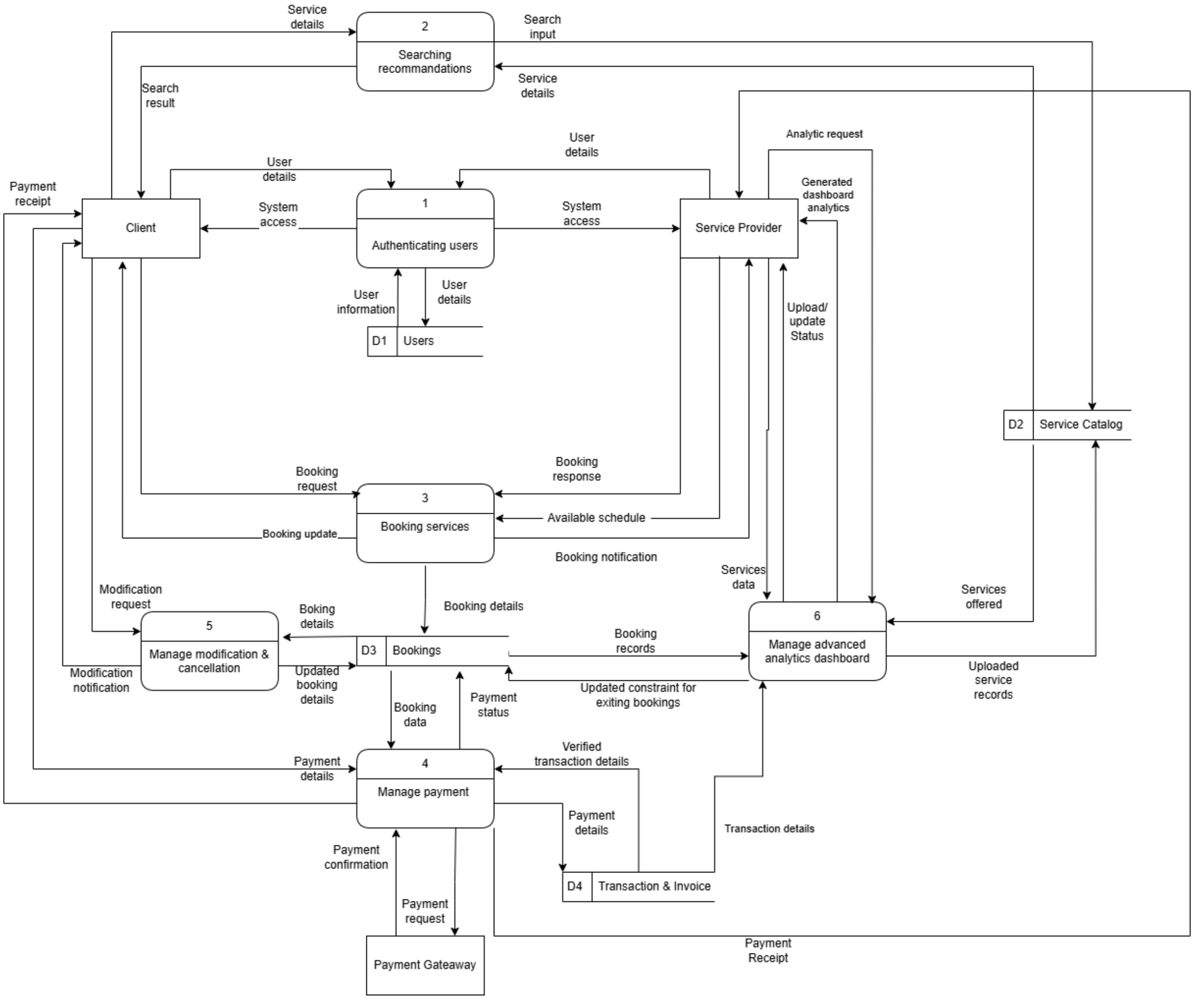

Figure 6.1.2 Diagram 0 of TO-BE system

Child diagram:

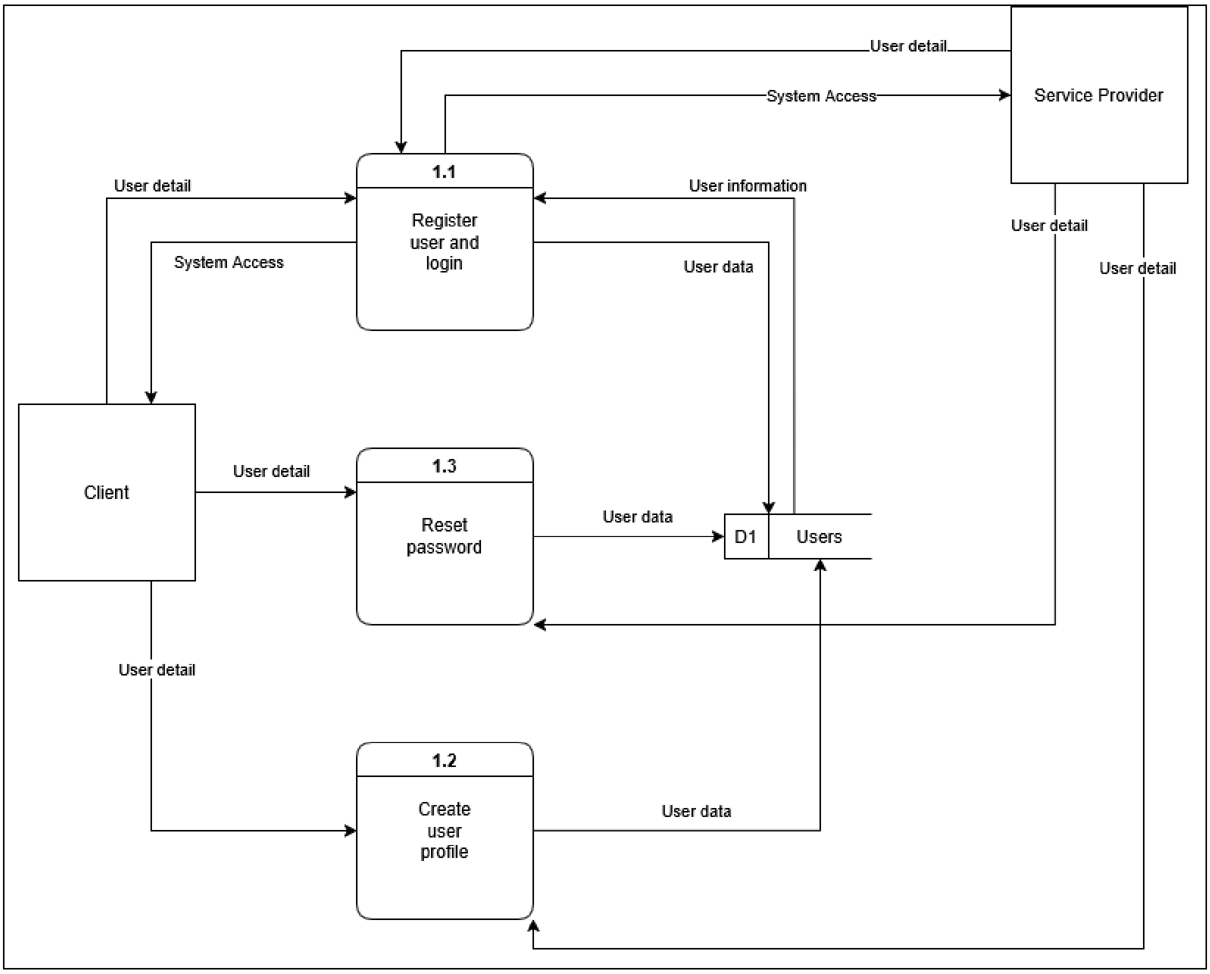

Figure 6.1.3 Child diagram of Process 1

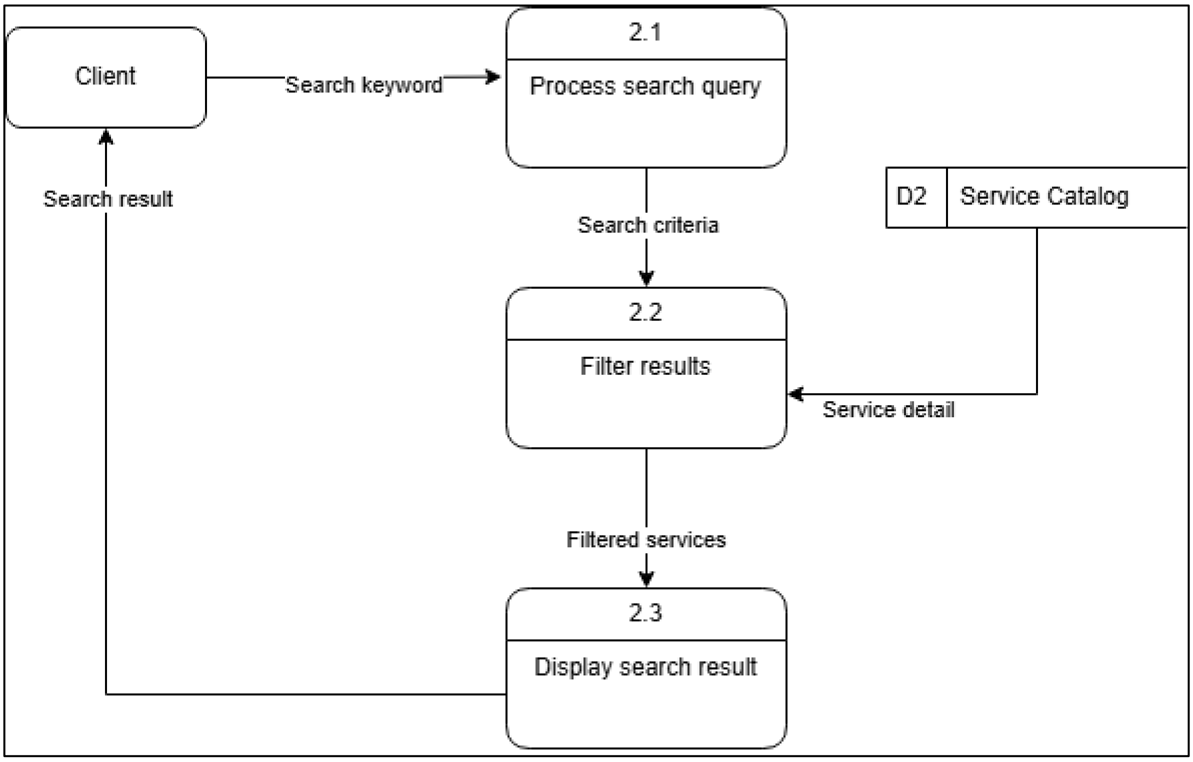

Figure 6.1.4 Child diagram of Process 2

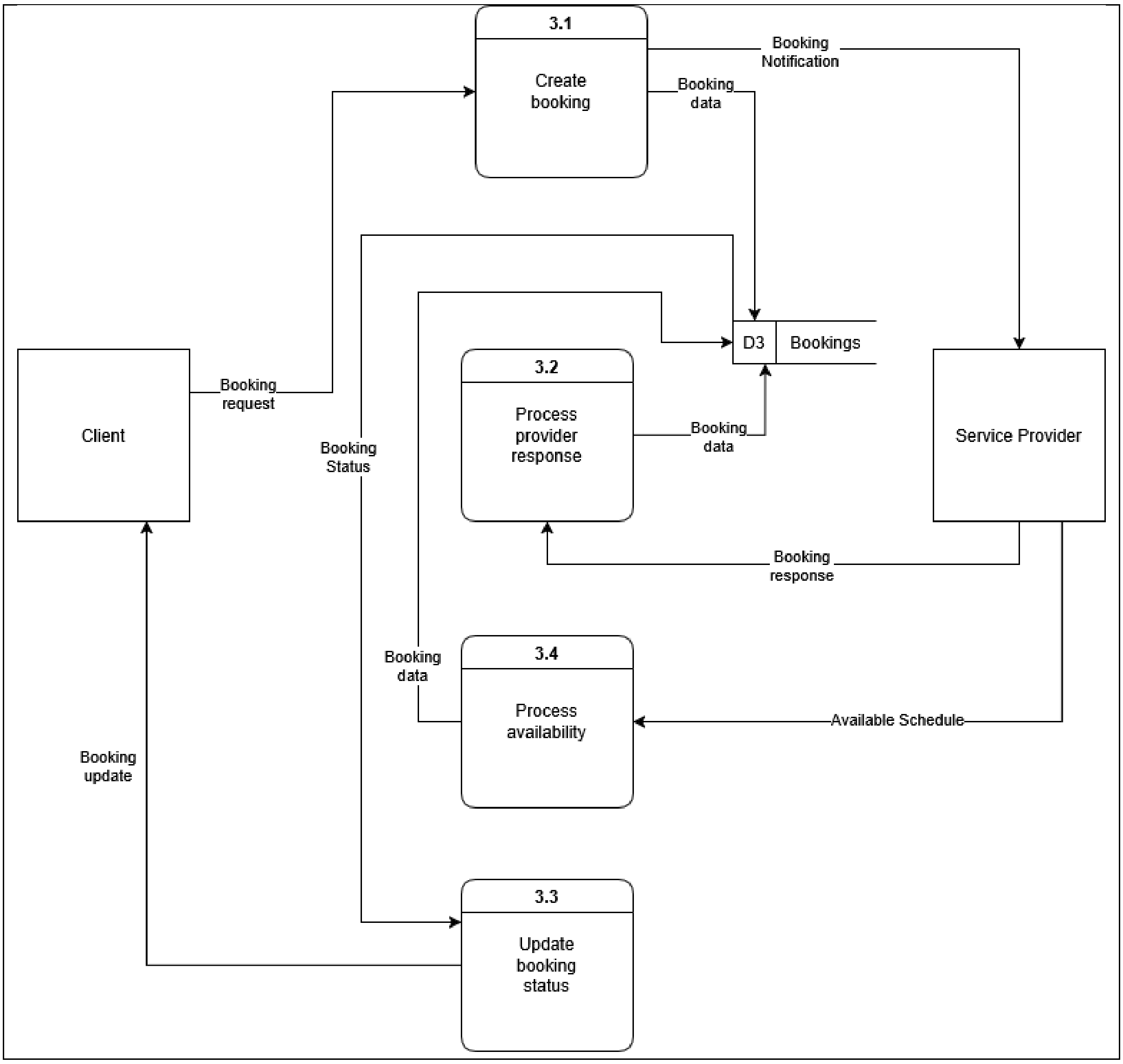

Figure 6.1.5 Child diagram of Process 3

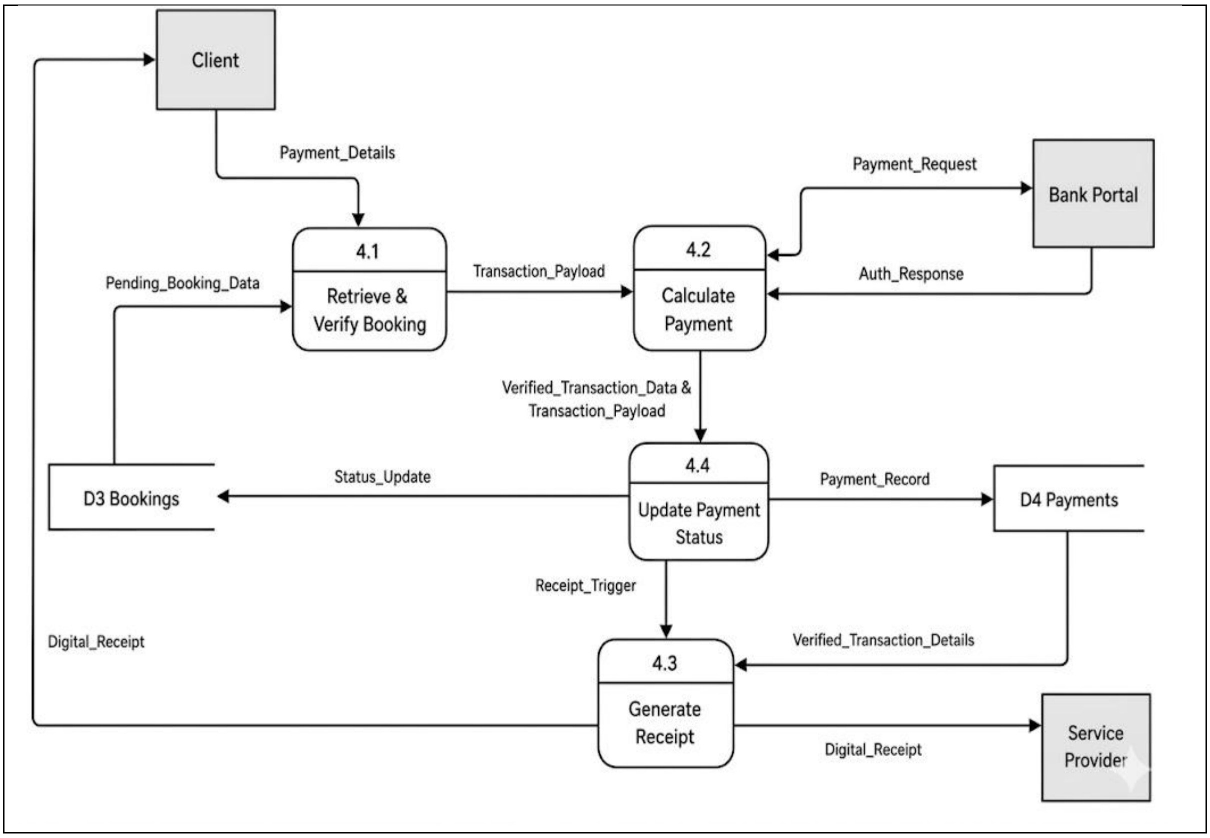

Figure 6.1.6 Child diagram of Process 4

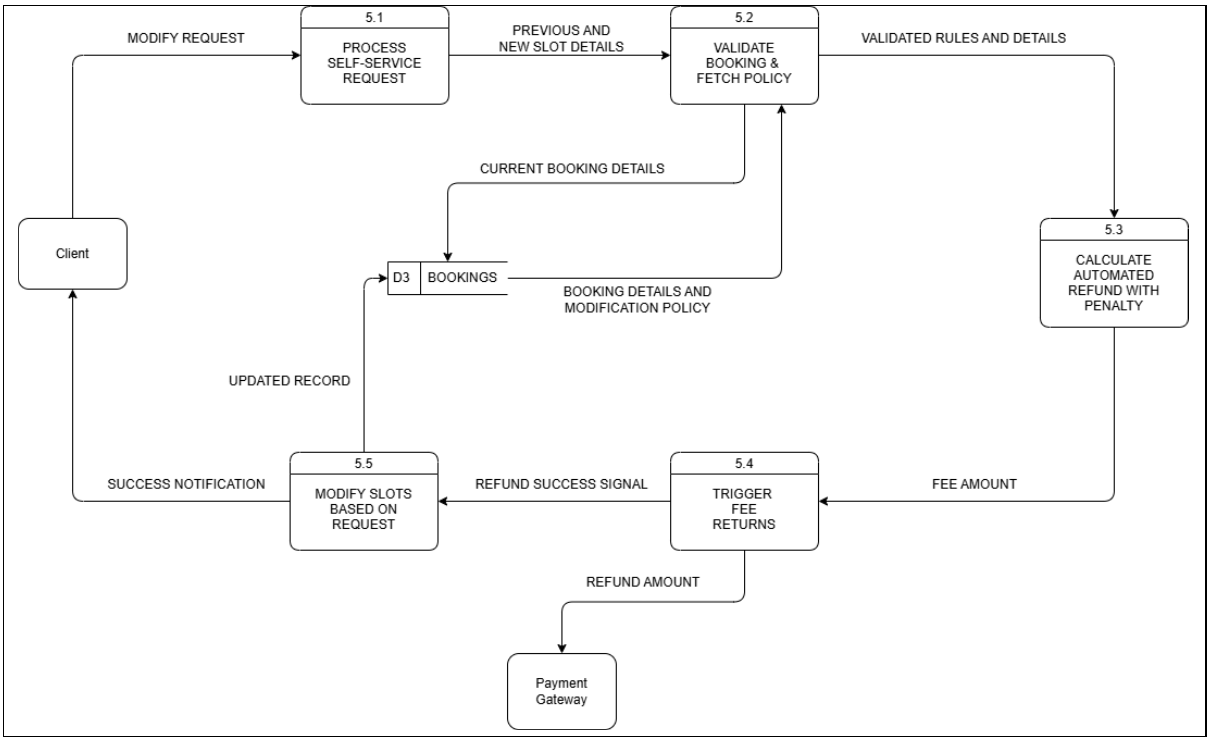

Figure 6.1.7 Child diagram of Process 5

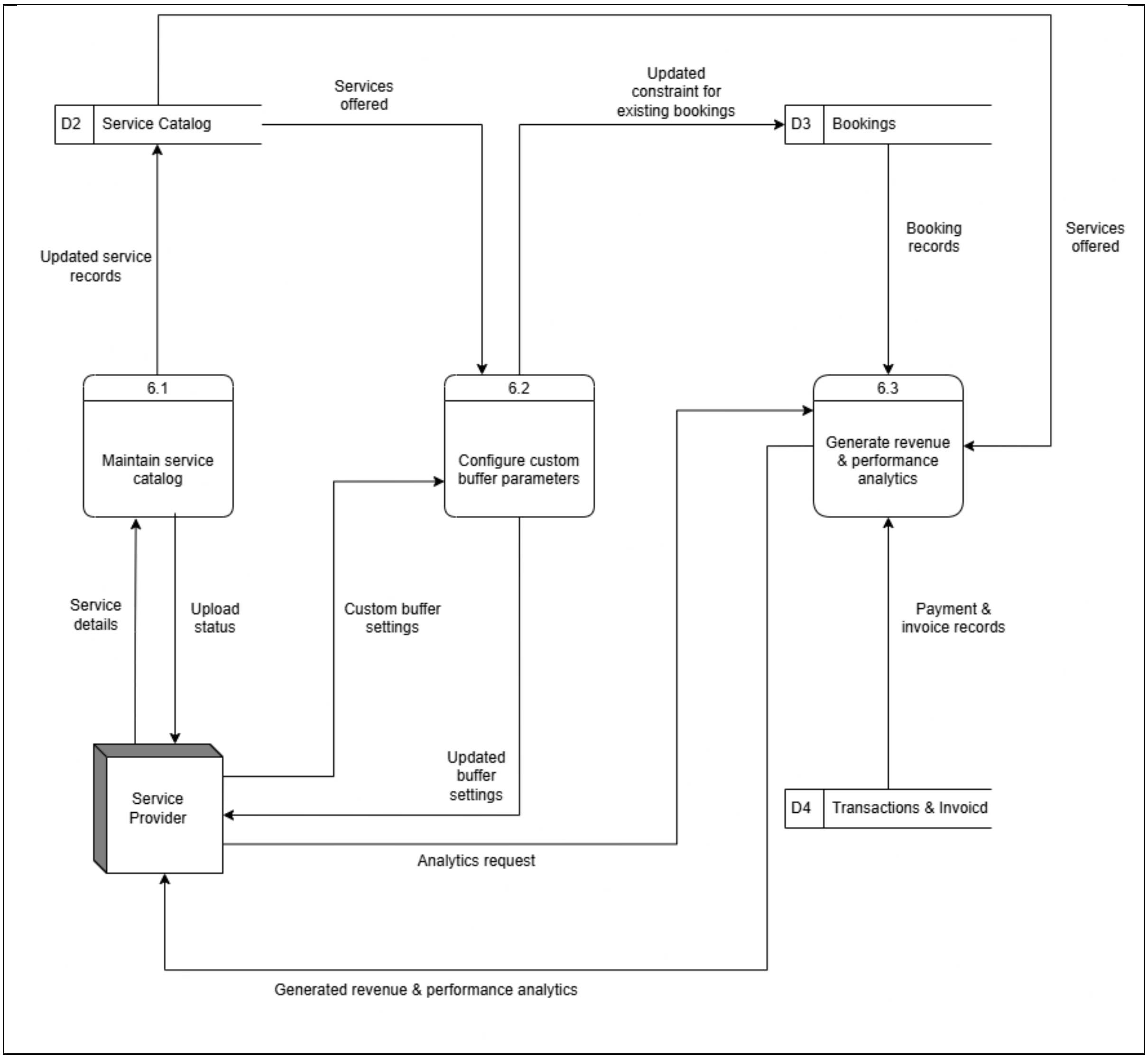

Figure 6.1.8 Child diagram of Process 6

6.2 Process Specification

Number: 1.1
Name: Register user and login
Description:
Receive user details from the Client or Service Provider to either authenticate an existing user or register a new one. Update the user database and grant system access upon successful authentication or registration.
Input Data Flow:

- User detail from Client
- User detail from Service Provider
- User data from D1 Users

Output Data Flow:

- System Access to Client
- System Access to Service Provider
- User information to D1 Users

Type of Process:

- Online

Subprogram/Function Name: User_Auth_And_Registration
Process Logic:

```
READ User detail from Client OR Service Provider
IF the request is for a new user registration THEN
   Validate User detail format
   IF the User detail is valid THEN
      DO write User information to D1 Users
      Move System Access authorization to Client OR Service Provider
   ELSE
      Reject registration request
   ENDIF
ELSE IF the request is for an existing user login THEN
   READ User data from D1 Users
   IF the provided User detail matches the saved User data THEN
      Move System Access authorization to Client OR Service Provider
   ELSE
      Reject login request
   ENDIF
ENDIF
```

Number: 1.2
Name: Create user profile
Description:
Receive user details from the Client or Service Provider to generate or update a user's profile information. Save the finalized user data into the Users database.

Input Data Flow:

- User detail from Client
- User detail from Service Provider
- User data from D1 Users

Output Data Flow:

- User data to D1 Users

Type of Process:

- Online

Subprogram/Function Name: User_Profile_Creation

Process Logic:

READ User detail from Client OR Service Provider
READ User data from D1 Users to check for existing profile
IF an existing profile is NOT found THEN
   Validate User detail for completeness
   IF the User detail is complete and valid THEN
      Format User detail into new User data record
      DO write User data to D1 Users
   ELSE
      Reject profile creation request and prompt for missing details
   ENDIF
ELSE
   Merge new User detail with existing User data
   DO write updated User data to D1 Users
ENDIF

Number: 1.3

Name: Reset password

Description:

Receive a password reset request containing user details from the Client or Service Provider. Verify the details against the database and update the user's record with a new password.

Input Data Flow:

- User detail from Client
- User detail from Service Provider
- User data from D1 Users (for verification)

Output Data Flow:

- User data to D1 Users

Type of Process:

- Online

Subprogram/Function Name: User_Password_Reset

Process Logic:

READ User detail (reset request) from Client OR Service Provider
READ User data from D1 Users

IF the provided User detail matches the saved User data for identity verification THEN
    Prompt Client OR Service Provider for new password
    Validate new password against security policies
    IF new password is valid THEN
        Update User data record with new password
        DO write updated User data to D1 Users
    ELSE
        Reject new password and prompt to try again
    ENDIF
ELSE
    Reject password reset request due to invalid identity
ENDIF

Figure 6.2.1 Structured English of Process 1

Number: 2.1
Name: Process search query
Description:
Receive keyword from Client and format the keyword into a query parameter that is suitable for processing
Input Data Flow:
- Search keyword

Output Data Flow:
- Formatted query parameters

Type of Process:
- Online

Subprogram/Function Name: Format_Search_Query
Process Logic:
READ searched keyword from Client
VALIDATE keyword to into query parameter
IF keyword is valid THEN
    SEND formatted query parameters to search API
ELSE
    DISPLAY error message
ENDIF

Number: 2.2
Name: Filter results
Description:
Use the formatted parameters to find services in D2 Service Catalog
Input Data Flow:
- Formatted query parameters

Output Data Flow:

- Filtered services listing

Type of Process:

- Online

Subprogram/Function Name: Fetch_Filtered_Services

Process Logic:

RECEIVE formatted query parameters
READ D2 Service Catalog
SEARCH database for matching service
IF matching service are found THEN
  COMPILE data into Filtered Service listing
ELSE
  GENERATE empty data with “no results” message
ENDIF
SEND filtered services listing to Render Result

Number: 2.3

Name: Display search result

Description:

Display the data received to client

Input Data Flow:

- Filtered service listing

Output Data Flow:

- Rendered result

Type of Process:

- Online

Subprogram/Function Name: Display_Search_Results

Process Logic:

RECEIVE filtered service listing
IF listing is not empty THEN
  DISPLAY data in the list
ELSE
  DISPLAY “no results” message
ENDIF
OUTPUT rendered result to Client

Figure 6.2.2 Structured English of Process 2

Number: 3.1

Name: Create booking

Description:

Receives a booking request from a client, validates the details, creates a new booking record in the database with an initial status, and alerts the service provider.

Input Data Flow:

- Booking request from Client

Output Data Flow:

- Booking data to D3 Bookings
- Booking Notification to Service Provider

Type of Process:

- Online

Subprogram/Function Name: Create_New_Booking
Process Logic:

```
READ Booking request from Client
Validate booking details (dates, service type, client information)
IF booking details are valid THEN
   Set initial booking status to 'Pending'
   Format request details into Booking data record
   DO write Booking data to D3 Bookings
   Generate Booking Notification
   Move Booking Notification to Service Provider
ELSE
   Reject Booking request and send error update to Client
ENDIF
```

Number: 3.2
Name: Process provider response
Description:
Receives the acceptance or rejection response from the service provider regarding a pending booking, and updates the database record accordingly.
Input Data Flow:

- Booking response from Service Provider

Output Data Flow:

- Booking data to D3 Bookings

Type of Process:

- Online

Subprogram/Function Name: Process_Provider_Response
Process Logic:

```
READ Booking response from Service Provider
Locate matching booking record in D3 Bookings using the booking ID
IF booking record is found THEN
   IF Booking response is 'Approved' THEN
      Update booking status to 'Confirmed'
   ELSE IF Booking response is 'Rejected' THEN
      Update booking status to 'Cancelled'
   ENDIF
   DO write updated Booking data to D3 Bookings
ELSE
   Log error: 'Booking record not found for the received response'
```

ENDIF

Number: 3.3
Name: Update booking status
Description:
Monitors or handles status updates from the bookings data store and passes a booking update notification to the client to keep them informed.
Input Data Flow:
- Booking Status from D3 Bookings

Output Data Flow:
- Booking update to Client

Type of Process:
- Online

Subprogram/Function Name: Send_Booking_Update
Process Logic:
READ Booking Status change from D3 Bookings
Extract relevant booking details (Status, Date, Provider Info)
Construct Booking update message
Move Booking update to Client via preferred notification channel

Number: 3.4
Name: Process availability
Description:
Processes schedule updates provided by the service provider, cross-references them against existing booking records to prevent overlaps, and updates the bookings availability data.
Input Data Flow:
- Available Schedule from Service Provider
- Booking data from D3 Bookings

Output Data Flow:
- Booking data (Availability updates) to D3 Bookings

Type of Process:
- Online

Subprogram/Function Name: Process_Schedule_Availability
Process Logic:
READ Available Schedule from Service Provider
READ existing Booking data from D3 Bookings
Compare incoming Available Schedule with existing confirmed bookings
IF a conflict is detected between a confirmed booking and the schedule change THEN
    Flag schedule conflict
    Notify Service Provider of conflicting booking slot
ELSE

Format schedule blocks into availability parameters
DO write updated availability details into Booking data within D3 Bookings
ENDIF

Figure 6.2.3 Structured English of Process 3

Process Number: 4.1
Process Name: Retrieve & Verify Booking
Description: This process retrieves pending booking parameters from the database to compute the final transaction cost and prepare a payment payload.
Input Data Flow: Pending_Booking_Data
Output Data Flow: Transaction_Payload
Type of Process: Online
Subprocess: None
Logic / Structured English:

```
BEGIN Retrieve & Verify Booking
READ Pending_Booking_Data FROM D3 Bookings

IF Pending_Booking_Data is found THEN
   EXTRACT hourly_rate AND session_duration
   EXTRACT Client_ID AND Service_Provider_ID

   COMPUTE session_cost = hourly_rate * session_duration

   IF AI_dynamic_pricing_factor exists THEN
     COMPUTE final_calculated_total = session_cost * AI_dynamic_pricing_factor
   ELSE
     COMPUTE final_calculated_total = session_cost
   END IF

 FORMAT Transaction_Payload using final_calculated_total AND Pending_Booking_Data
   PASS Transaction_Payload TO Process 4.2
ELSE
   RETURN "Error: Booking Record Not Found"
END IF
END Retrieve & Verify Booking
```

Process Number: 4.2
Process Name: Calculate Payment
Description: This process transmits payment parameters and calculated total to the Bank Portal for authorization.
Input Data Flow: Transaction_Payload, Payment_Details, Auth_Response

Output Data Flow: Payment_Request, Verified_Transaction_Data, Transaction_Payload
Type of Process: Online
Subprocess: None
Logic / Structured English:

```
BEGIN Calculate Payment
RECEIVE Transaction_Payload FROM Process 4.1
RECEIVE Payment_Details FROM Client

FORMAT Payment_Request using Transaction_Payload AND Payment_Details
TRANSMIT Payment_Request TO Bank Portal
WAIT FOR Auth_Response FROM Bank Portal

IF Auth_Response is "Approved" THEN
   GENERATE Verified_Transaction_Data
   PASS Verified_Transaction_Data AND Transaction_Payload TO Process 4.4
ELSE
   RETURN "Error: Payment Authorization Failed" TO Client
   TERMINATE Process
END IF
END Calculate Payment
```

Process Number: 4.3
Process Name: Generate Receipt
Description: This process generates a digital receipt from finalized transaction data and sends it to both parties.
Input Data Flow: Receipt_Trigger, Verified_Transaction_Details
Output Data Flow: Digital_Receipt
Type of Process: Online
Subprocess: None
Logic / Structured English:

```
BEGIN Generate Receipt
  RECEIVE Receipt_Trigger FROM Process 4.4
  READ Verified_Transaction_Details FROM D4 Payments

  FORMAT Digital_Receipt using Verified_Transaction_Details
  INCLUDE date, time, Service_Provider_ID, Client_ID, final_calculated_total

  SEND Digital_Receipt TO Client
  SEND Digital_Receipt TO Service Provider
END Generate Receipt
```

Process Number: 4.4
Process Name: Update Payment Status
Description: This process writes the new transaction data into the payments database and updates the master booking status from pending to confirmed.
Input Data Flow: Verified_Transaction_Data, Transaction_Payload
Output Data Flow: Payment_Record, Status_Update, Receipt_Trigger
Type of Process: Online
Subprocess: None
Logic / Structured English:

```
BEGIN Update Payment Status
  RECEIVE Verified_Transaction_Data AND Transaction_Payload FROM Process
4.2

  CREATE Payment_Record IN D4 Payments using Verified_Transaction_Data
  UPDATE booking_status = "Confirmed" IN D3 Bookings for corresponding
Transaction_Payload

  IF database updates are successful THEN
    PASS Receipt_Trigger TO Process 4.3
  ELSE
    LOG "Database Write Error"
   ALERT System Administrator
 END IF
END Update Payment Status
```

Figure 6.2.4 Structured English of Process 4

Number: 5.1
Name: Process Self-Service Request
Description:
This process triggers when users want to reschedule or cancel their bookings, where users click on the corresponding button based on their demand. The system will then retrieve the previous and new slot details.
Input Data Flow:
- Modify Request from Client

Output Data Flow:
- Previous and New Slot Details to Process 5.2

Type of Process:
- Online

Subprogram/Function Name: None
Process Logic:

BEGIN Process Self-Service Request

RECEIVE Modify Request FROM Client
EXTRACT Previous and New Slot Details
SEND Previous and New Slot Details to Process 5.2

END Process Self-Service Request

Number: 5.2
Name: Validate Booking & Fetch Policy
Description:
This process will cross-check the booking details with the reschedule policy to validate the modification request.
Input Data Flow:

- Previous and New Slot Details from Process 5.1
- Booking Details and Modification Policy from D3 Bookings

Output Data Flow:

- Validated Rules and Details to Process 5.3
- Current Booking Details to D3 Bookings

Type of Process:

- Online

Subprogram/Function Name: None
Process Logic:

BEGIN Validate Booking & Fetch Policy

RECEIVE Previous and New Slot Details
SEND Current Booking Details to D3 Bookings
VALIDATE the Modification Request
FETCH Booking Details and Modification Policy from D3 Bookings
SEND Validated Rules and Details to Process 5.3

END Validate Booking & Fetch Policy

Number: 5.3
Name: Calculate Automated Refund with Penalty
Description:
This process will calculate the refund automatically with the penalty (if applicable) to be paid by the users.
Input Data Flow:

- Validated Rules and Details from Process 5.2

Output Data Flow:

- Fee Amount to Process 5.4

Type of Process:

- Online

Subprogram/Function Name: None

Process Logic:

```
BEGIN Calculate Automated Refund with Penalty

   RECEIVE Validated Rules and Details
   COMPUTE financial adjustments based on policy
   CALCULATE Fee Amount
   SEND Fee Amount to Process 5.4

END Calculate Automated Refund with Penalty
```

Number: 5.4

Name: Trigger Fee Returns

Description:

This process will initiate a refund after penalty deducting to the payment gateway. If the process is successful, the system will send a successful signal.

Input Data Flow:

- Fee Amount from Process 5.3

Output Data Flow:

- Refund Success Signal to Process 5.5
- Refund Amount to Payment Gateway

Type of Process:

- Online

Subprogram/Function Name: None

Process Logic:

```
BEGIN Trigger Fee Returns

   RECEIVE Fee Amount
   SEND Refund Amount to Payment Gateway
   WAIT for confirmation

   IF refund is successful THEN
      SEND Refund Success Signal to Process 5.5
   END IF

END Trigger Fee Returns
```

Number: 5.5

Name: Modify Slots Based on Request

Description:
This process will alter the booking slot to the new requested slot or just cancel the booking, according to users' demands, then notify the users with success notification.
Input Data Flow:
- Refund Success Signal from Process 5.4

Output Data Flow:
- Success Notification to Client
- Updated Record to D3 Bookings

Type of Process:
- Online

Subprogram/Function Name: None
Process Logic:

BEGIN Modify Slots Based on Request

RECEIVE Refund Success Signal
SAVE Updated Record into D3 Bookings
SEND Success Notification to Client

END Modify Slots Based on Request

Figure 6.2.5 Structured English of Process 5

Number: 6.1
Name: Bulk Upload Inventory
Description:
This process triggers when Service Provider wants to upload bulk services data into the system using a PDF or CSV file. The system will then validate the data uploaded and then returns the upload status
Input Data Flow:
- Bulk Service Data from Service Provider

Output Data Flow:
- Validated Service Record to Service Catalog
- Upload Status to Service Provider

Type of Process:
- Online

Subprogram/Function Name: None
Process Logic:

BEGIN Bulk Upload Inventory
READ Bulk Service Data FROM Service Provider
VALIDATE uploaded CSV/Excel file format and data integrity

CHECK for duplicate services or missing required fields
IF data is valid THEN
CREATE new service records
WRITE bulk service records TO Service Catalog
SEND upload status success TO Service Provider
ELSE
SEND upload status error TO Service Provider
END IF
END Bulk Upload Inventory

Number: 6.2
Name: Configure Custom Buffers
Description:
This process triggers when Service Provider wants to update or add custom buffer configuration that will affect the constraint put on bookings. The system will then update the constraint if prompted and return AI-generated preview impact to Service Provider
Input Data Flow:
- List of Services Offered from Service Catalog
- Custom Buffer Settings from Service Provider

Output Data Flow:
- Updated constraint for existing bookings to Bookings
- AI-generated preview impact to Service Provider

Type of Process:
- Online

Subprogram/Function Name: None
Process Logic:

BEGIN Configure Custom Buffers
READ custom buffer settings FROM Service Provider
VALIDATE custom buffer parameters
UPDATE service buffer rules in Service Catalog
APPLY new availability constraints TO Bookings
SEND configuration confirmation and impact preview TO Service Provider
END Configure Custom Buffers

Number: 6.3
Name: Generate Revenue & Performance Analytics
Description:
This process triggers when Service Provider clicks on the Analytics Dashboard page, which contains data generated related to revenue and performance. The system will return a dashboard view, and Service Provider can choose to export the analytics as PDF/CSV.

Input Data Flow:

- Booking performance data FROM Bookings
- Revenue data FROM Transactions & Invoice
- Services offered data FROM Service Catalog
- Analytics request from Service Provider

Output Data Flow:

- Analytics Dashboard View to Service Provider
- Exported analytics in PDF/CSV to Service Provider

Type of Process:

- Online

Subprogram/Function Name: None

Process Logic:

```
BEGIN Generate Revenue & Performance Analytics
    READ revenue data FROM Transactions & Invoice
    READ booking performance data FROM Bookings
    READ services offered data FROM Service Catalog
    CALCULATE key metrics for analytics
    GENERATE dashboard visualizations and reports
    SEND Analytics Dashboard View TO Service Provider
    IF Service Provider clicks ‘Export’ THEN
      READ revenue data FROM Transactions & Invoice
      READ booking performance data FROM Bookings
      GENERATE formatted report in PDF or CSV
      SEND exported analytics in PDF/CSV TO Service Provider
    END IF
END Generate Revenue & Performance Analytics
```

Figure 6.2.6 Structured English of Process 6

7. Physical System Design

7.1 Physical DFD TO-BE system

Diagram 0

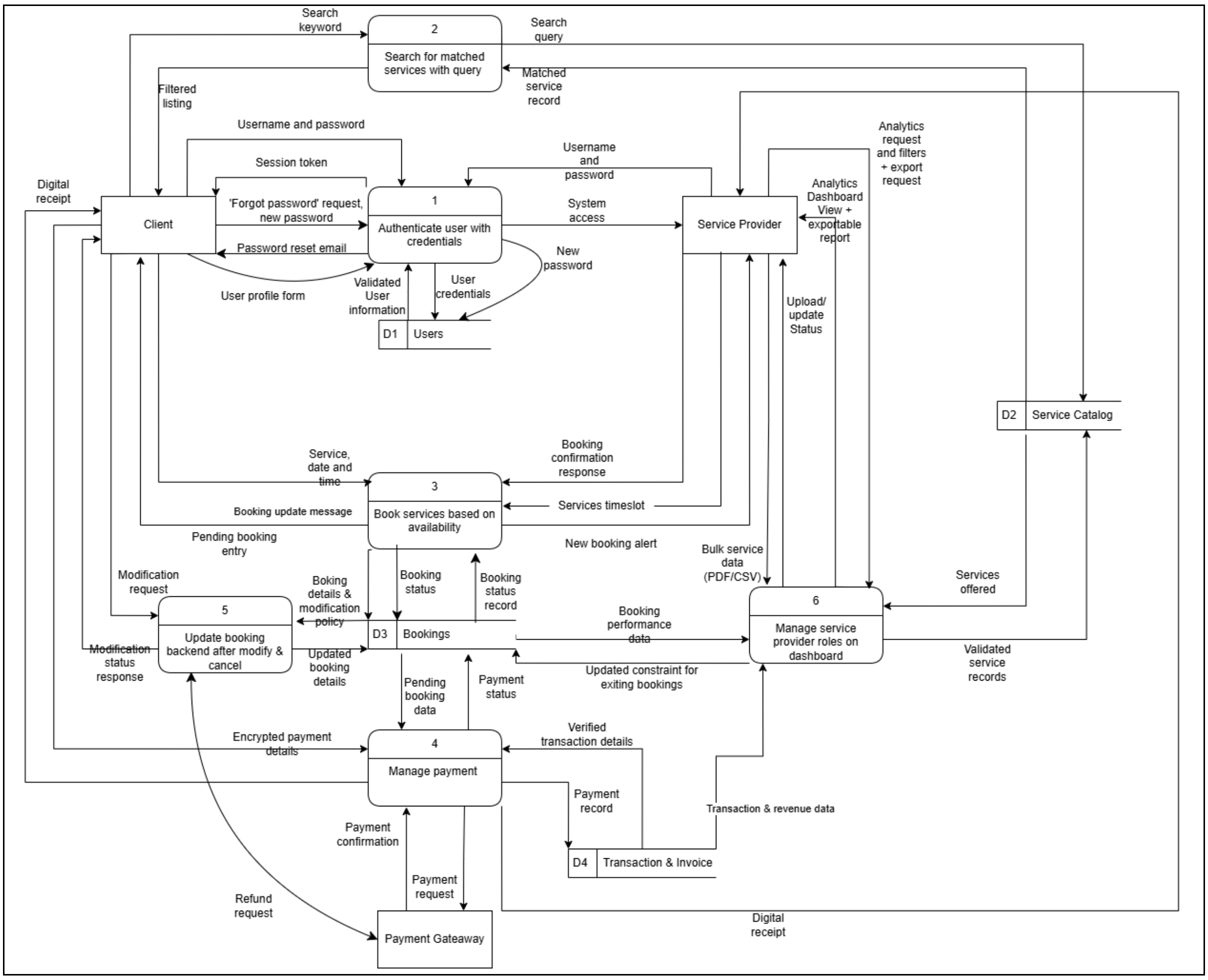

Figure 7.1.1 Physical DFD of diagram zero

Child diagram

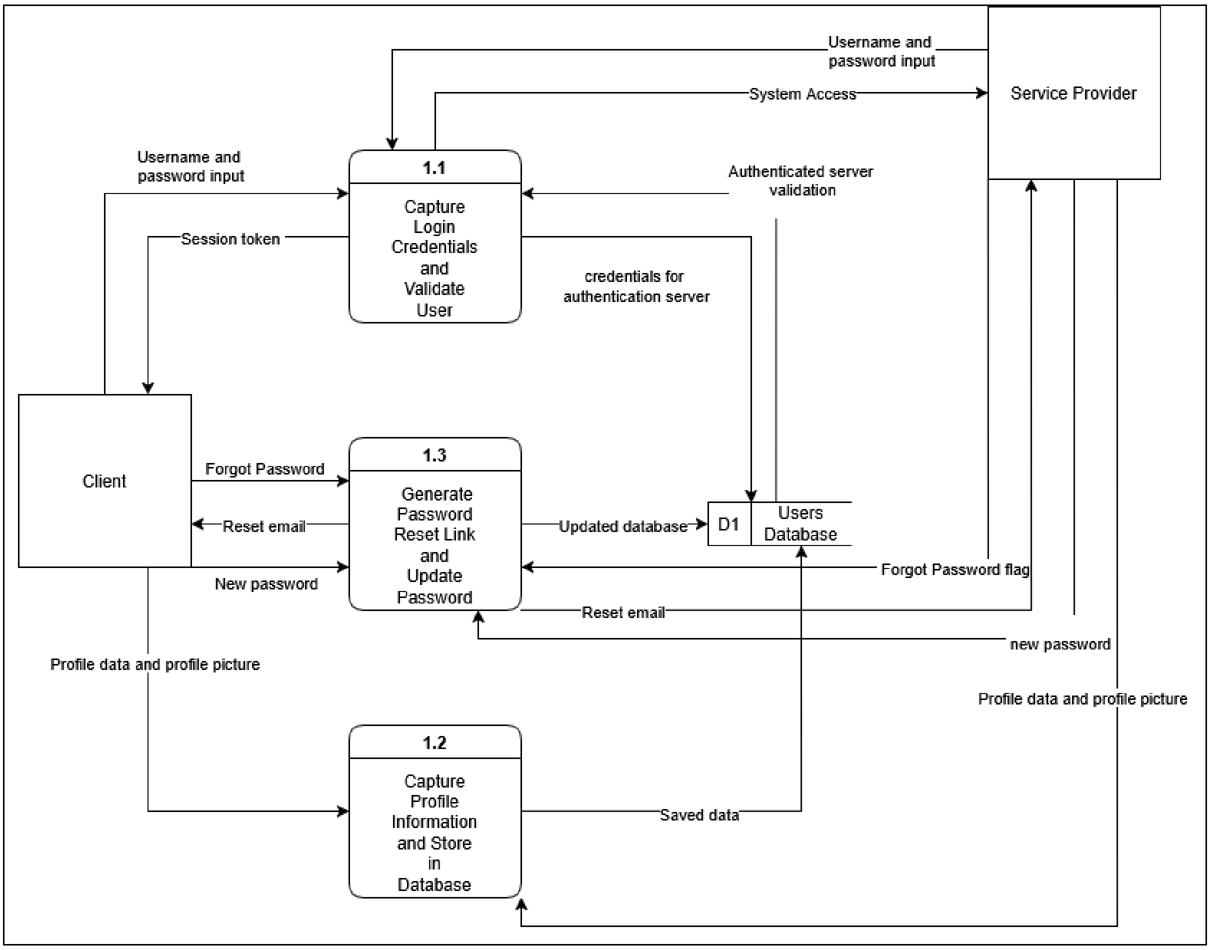

Figure 7.1.2 Physical DFD of Process 1

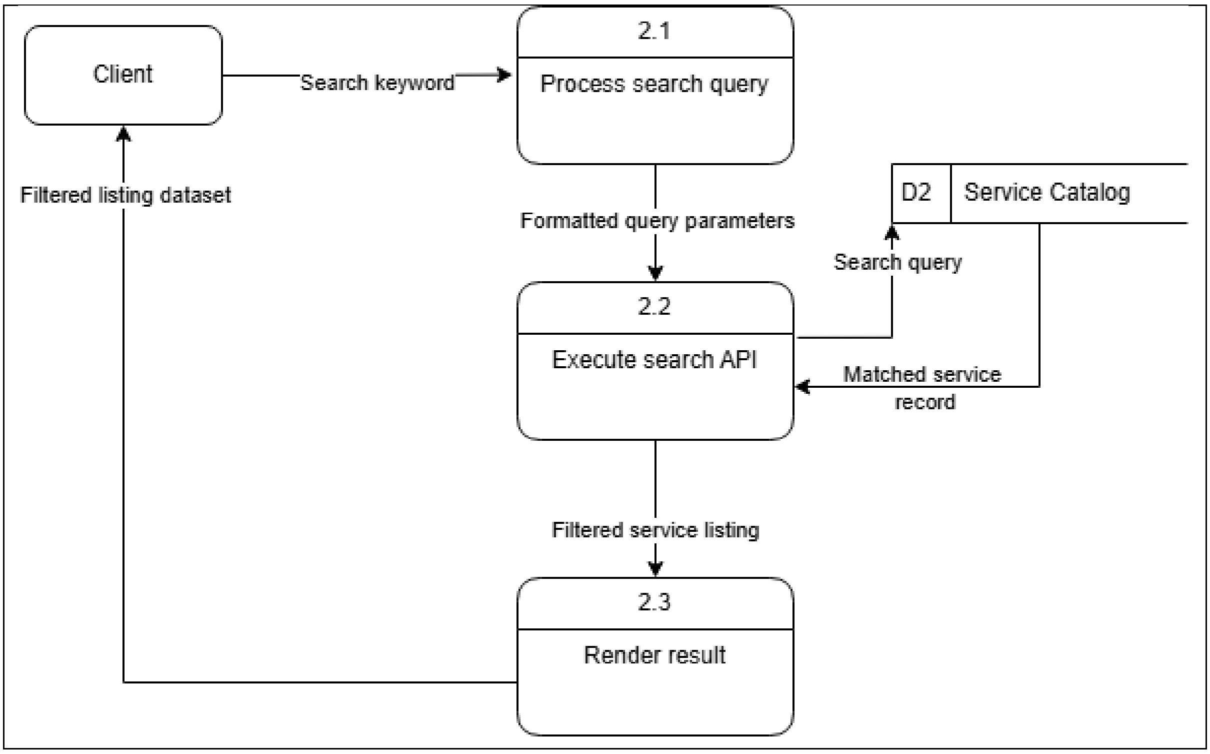

Figure 7.1.3 Physical DFD of Process 2

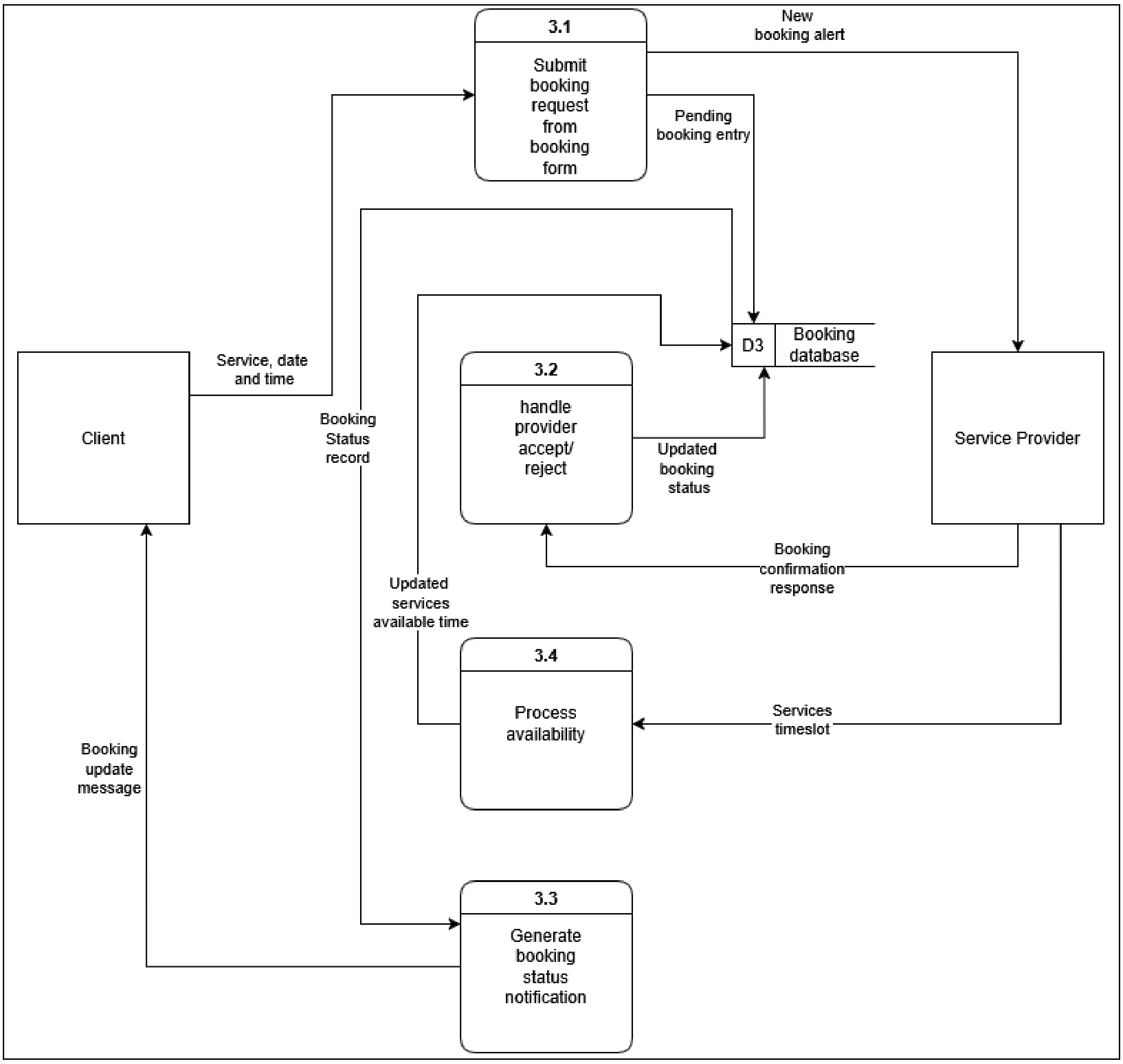

Figure 7.1.4 Physical DFD of Process 3

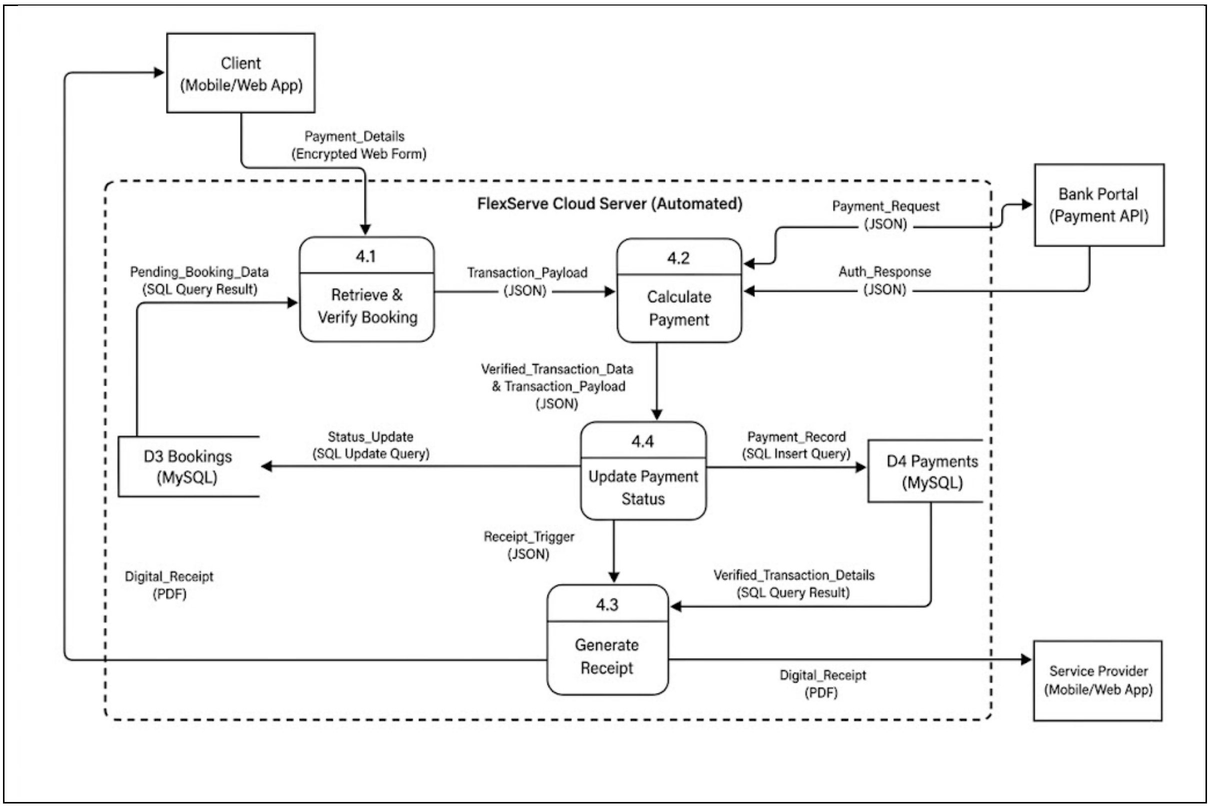

Figure 7.1.5 Physical DFD of Process 4

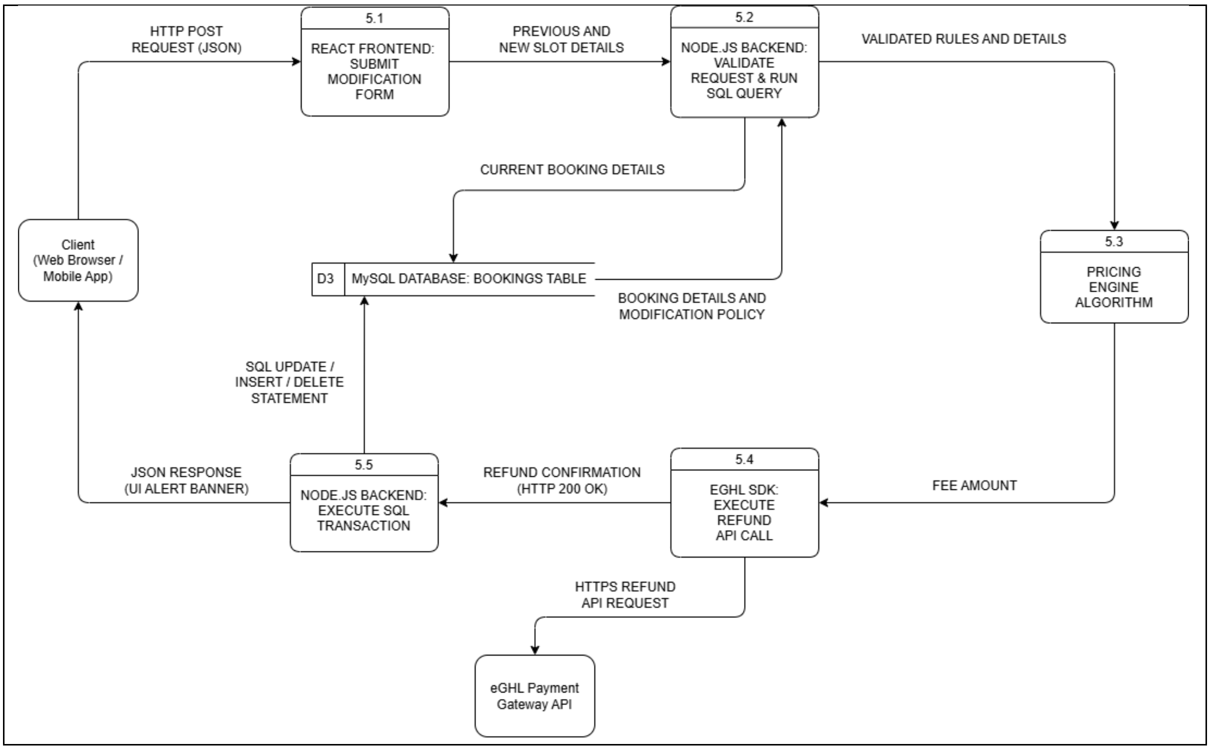

Figure 7.1.6 Physical DFD of Process 5

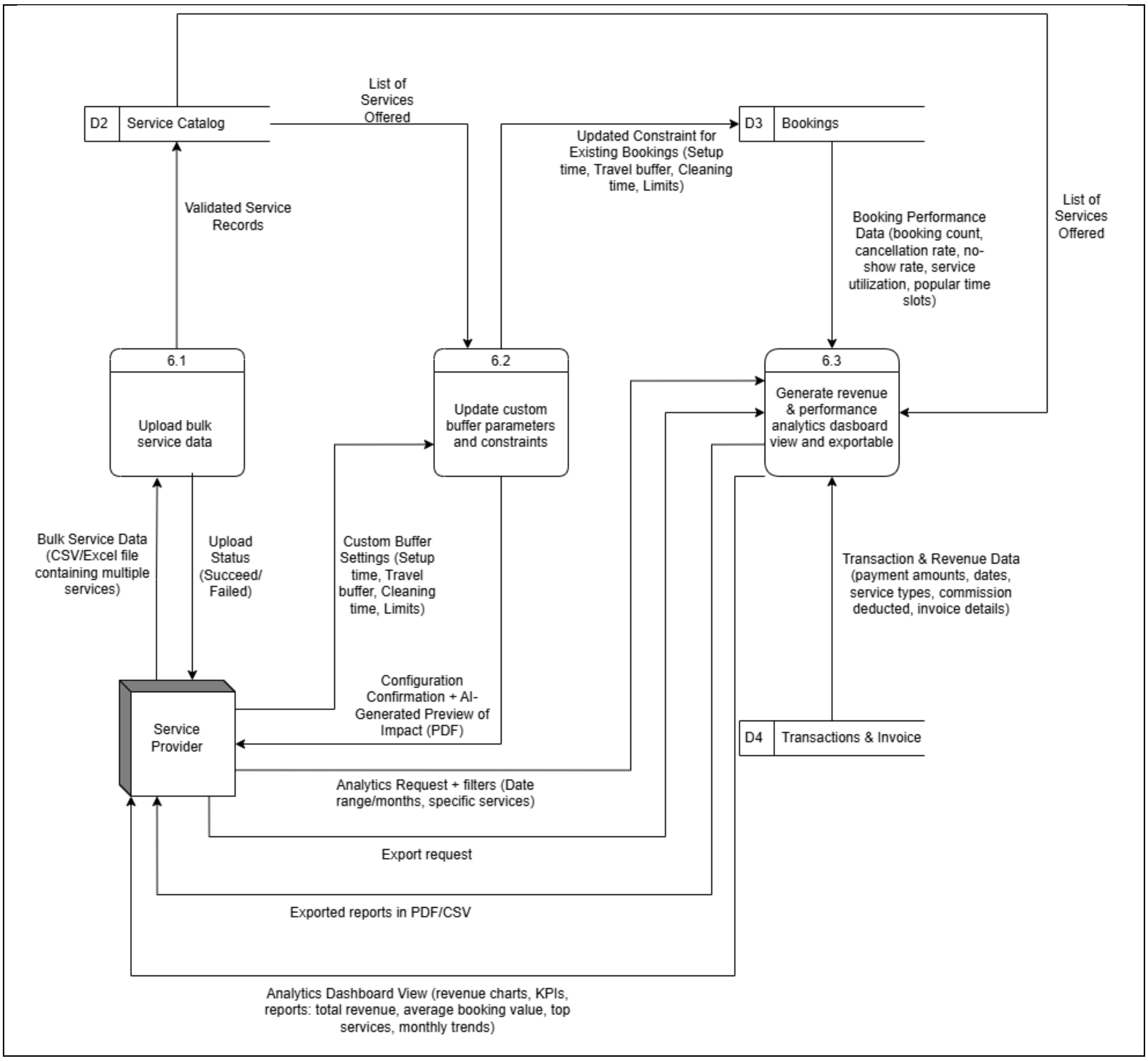

Figure 7.1.7 Physical DFD of Process 6

7.2 Partitioning

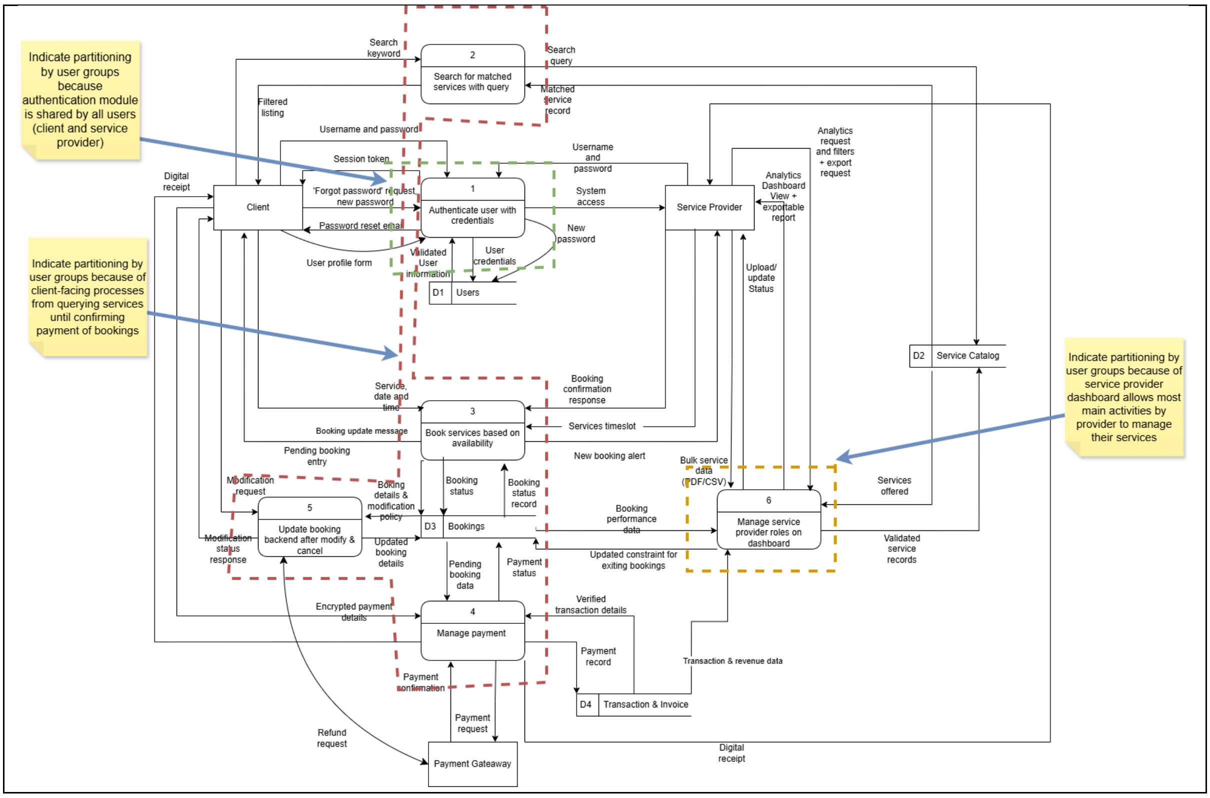

Figure 7.2.1 Partitioning Diagram 0

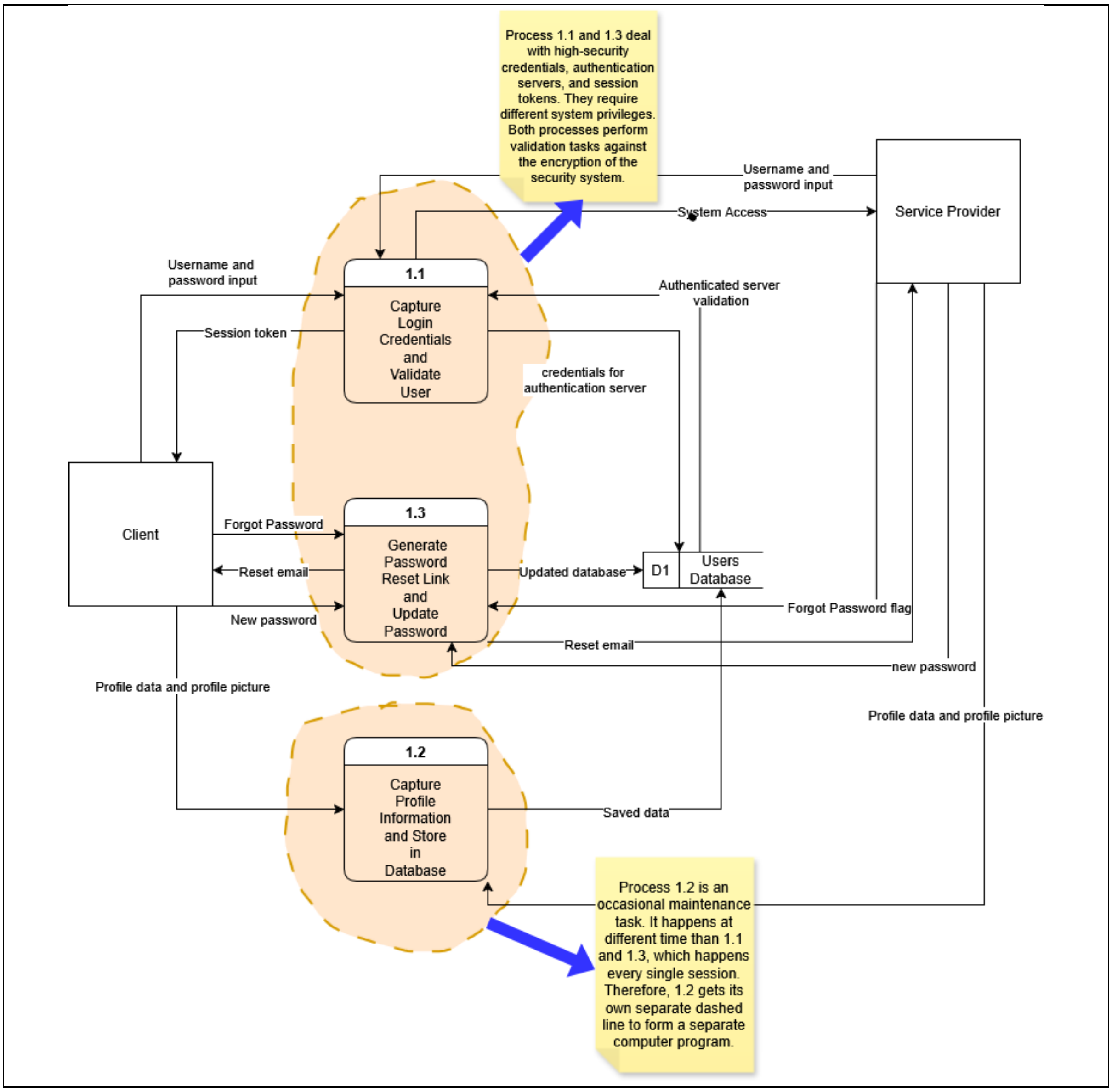

Figure 7.2.2 Partitioning Physical DFD Process 1

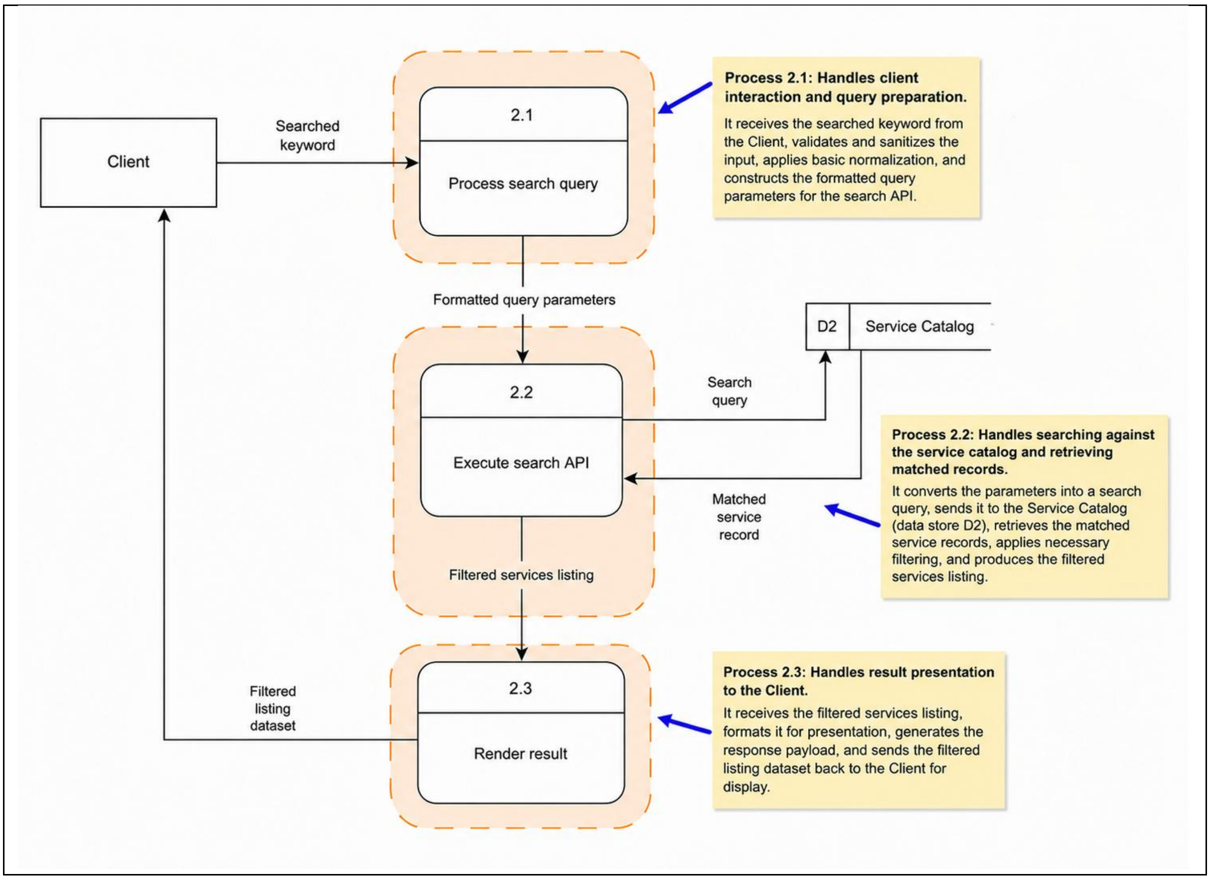

Figure 7.2.3 Partitioning Physical DFD Process 2

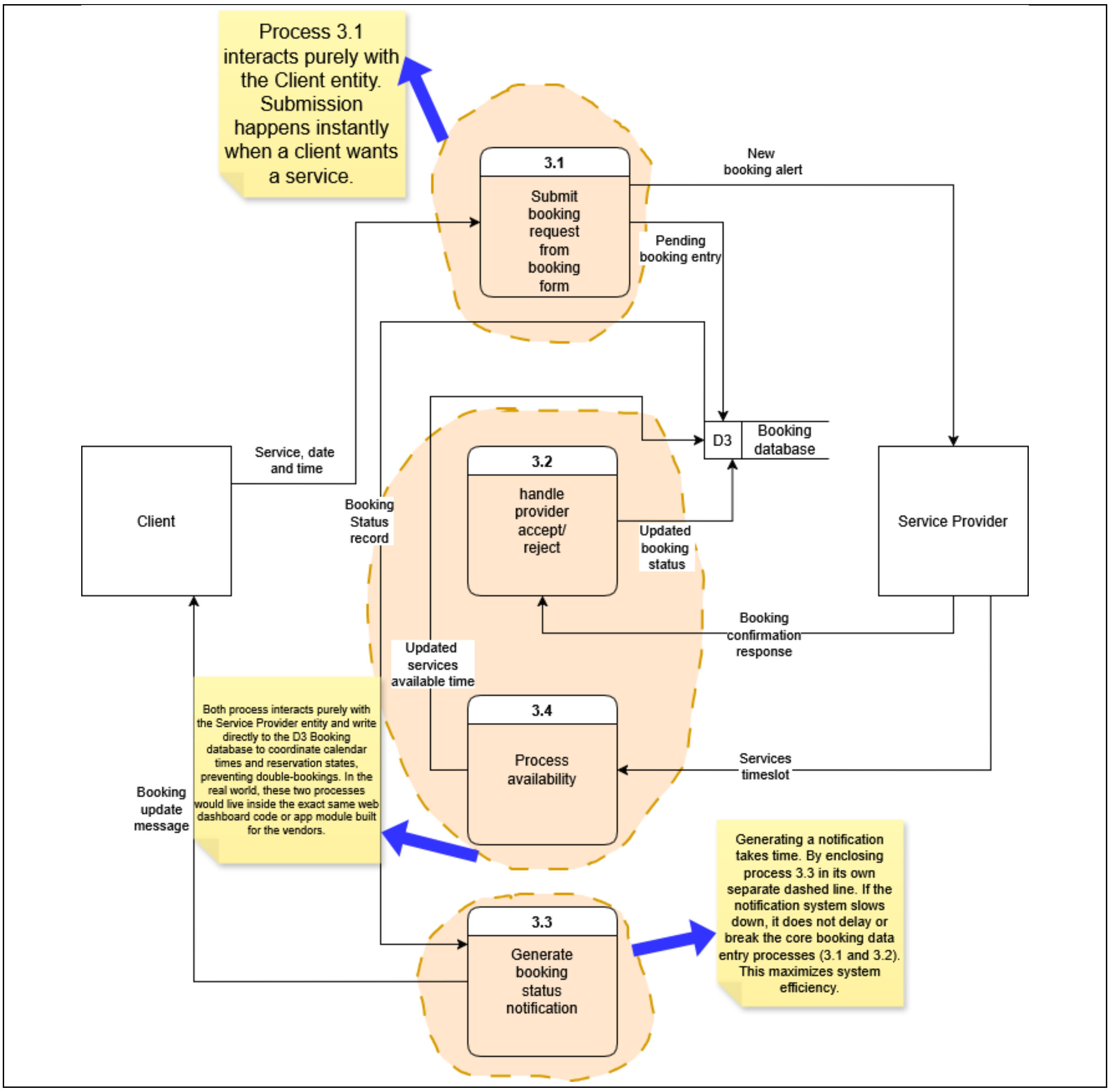

Figure 7.2.4 Partitioning Physical DFD Process 3

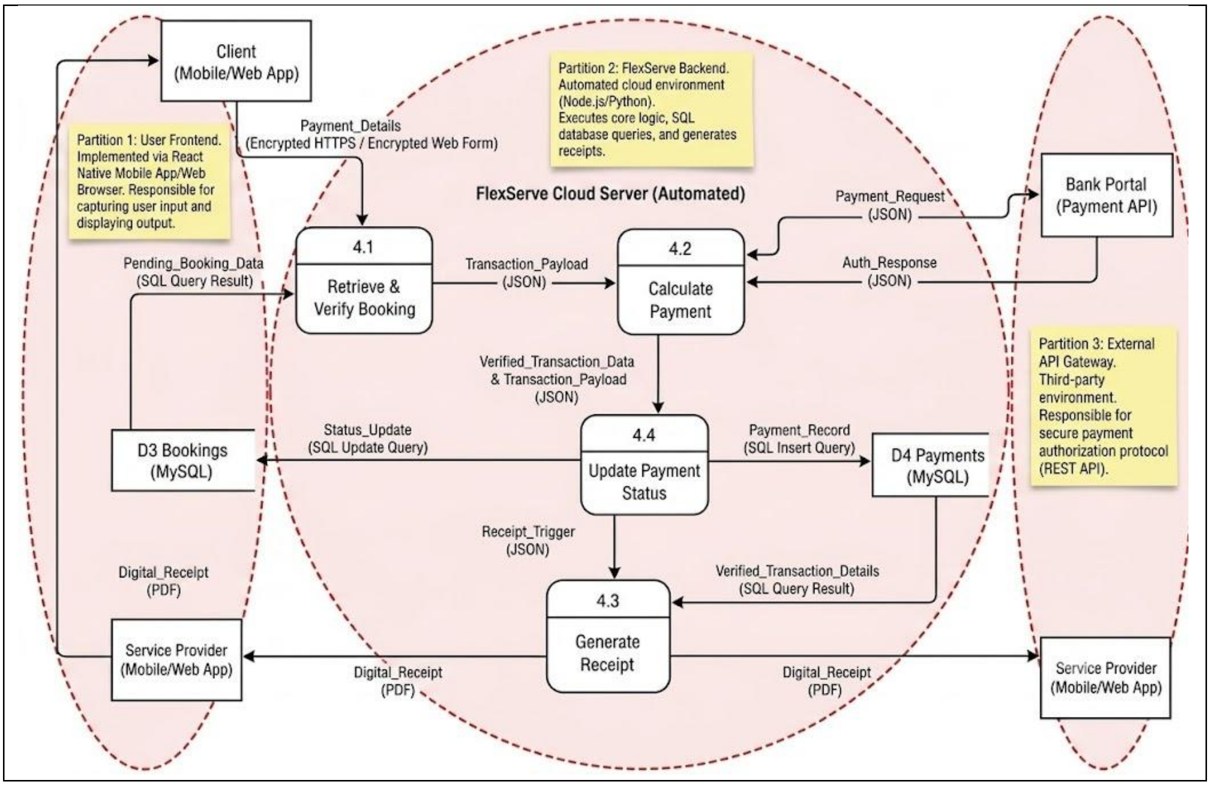

Figure 7.2.5 Partitioning Physical DFD Process 4

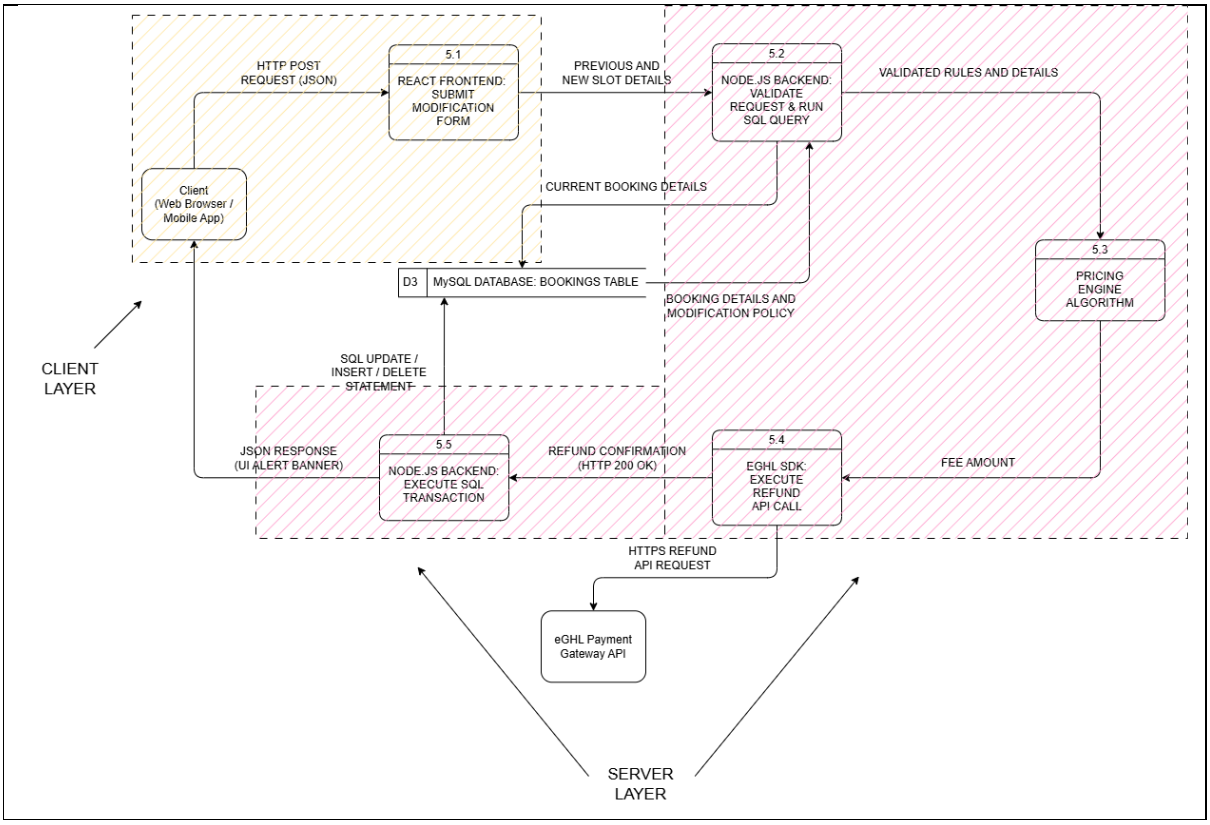

Figure 7.2.6 Partitioning Physical DFD Process 5

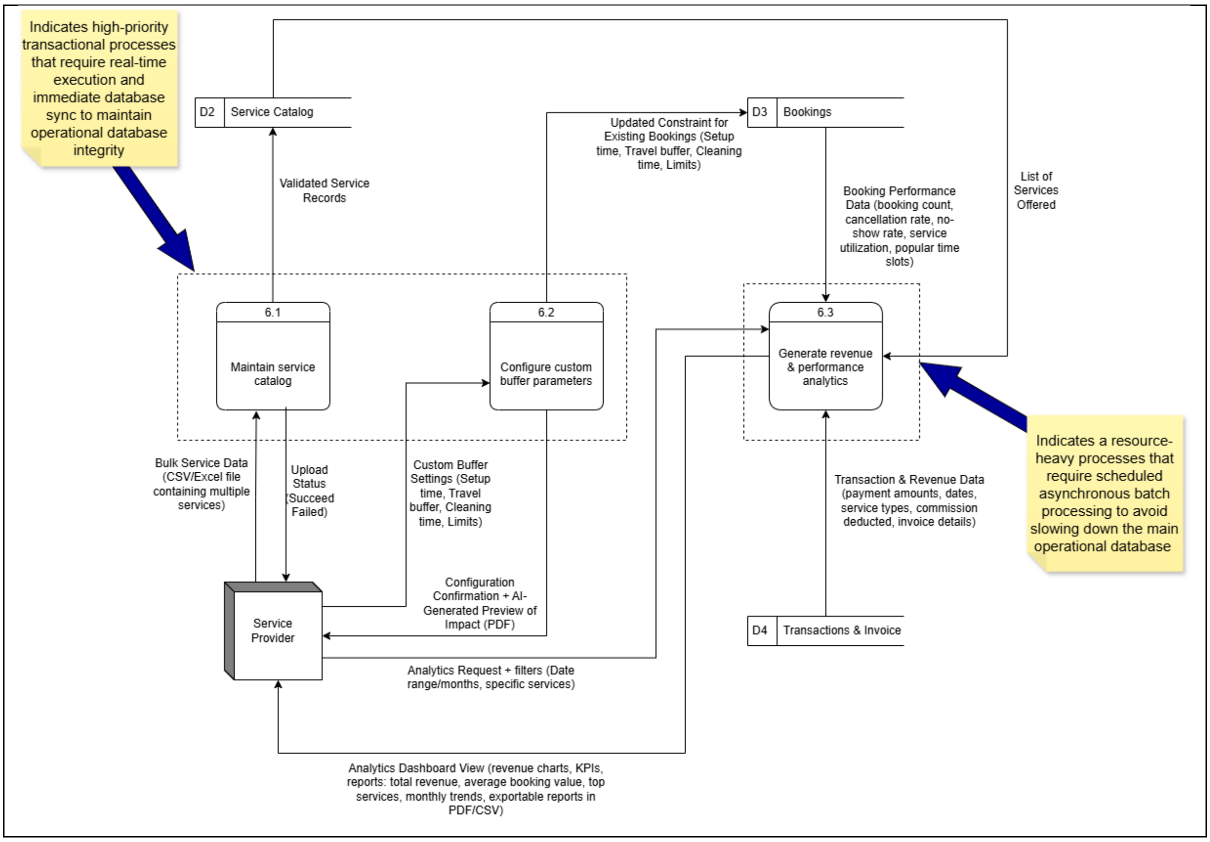

Figure 7.2.7 Partitioning Physical DFD Process 6

7.3 CRUD Matrix

Process Name	D1 Users
1.1 Register user and login	CR (Create, Read)
1.2 Create user profile	C (Create)
1.3 Reset password	U (Update)

Figure 7.3.1 CRUD Matrix of Process 1

Process Name	D2 Service Catalog
2.1 Process search query	-
2.2 Execute search API	R
2.3 Render result	-

Figure 7.3.2 CRUD Matrix of Process 2

Process Name	D3 Bookings
3.1 Create booking	C (Create)
3.2 Process provider response	U (Update)
3.3 Update booking status	R (Read)
3.4 Process availability	U (Update)

Figure 7.3.3 CRUD Matrix of Process 3

Process Name	D3 Bookings (MySQL)	D4 Payments (MySQL)
4.1 Retrieve & Verify Booking	R(Read)	-
4.2 Calculate Payment	-	-
4.3 Generate Receipt	R(Read)	R(Read)
4.4 Update Payment Status	U(Update)	C(Create)

Figure 7.3.4 CRUD Matrix of Process 4

Process Name	D3 Bookings	Client	Payment Gateway
5.1 Process Self-Service Request	-	R (Read)	-
5.2 Validate Booking & Fetch Policy	R (Read)	-	-
5.3 Calculate Automated Refund with Penalty	-	-	-
5.4 Trigger Fee Returns	-	-	U (Update)
5.5 Modify Slots Based on Request	U (Update)	U (Update)	-

Figure 7.3.5 CRUD Matrix of Process 5

Process Name	D2 Service Catalog	D3 Bookings	D4 Transactions & Invoice
6.1 Mantain Service Catalog	C (Create)	-	-
6.2 Configure Custom Buffer Parameters	R (Read)	U (Update)	-
6.3 Generate Revenue & Performance Analytics	R (Read)	R (Read)	R (Read)

Figure 7.3.6 CRUD Matrix of Process 6

7.4 Event Response Table (ERT)

Event	Source	Trigger	Activity	Response	Destination
User or Service Provider logs into the system	Client, Service Provider	User enters username and password	1.1 Capture Login Credentials and Validate User	Session token is generated, System access provided	Client, Service Provider
User or Service Provider updates profile	Client, Service Provider	User fills profile form and Upload profile picture	1.2 Create user profile	Save data into Users database	D1 Users Database
User or Service Provider requests password reset	Client, Service Provider	User clicks Forgot Password	1.3 Reset password	System sends reset email	Client, Service Provider
User or Service Provider submits new password	Client, Service Provider	User enters new password	1.3 Reset password	Update Password	Database is updated

Figure 7.4.1 ERT of Process 1

Event	Source	Trigger	Activity	Response	Destination
Client submits a search keyword	Client	User enters and submits search input	2.1 Process search query	Validates, formats and prepares query parameters	Process 2.2
Search request is ready for execution	Process 2.1	Formatted query parameters received	2.2 Execute search API	Convert parameter into a search query and retrieve matching	Process 2.3

				service record from D2	
Search results are available	Process 2.2	Filtered services listing received	2.3 Render result	Formats and prepares the result for display	Client

Figure 7.4.2 ERT of Process 2

Event	Source	Trigger	Activity	Response	Destination
Client initiates a booking	Client	Service, date and time	3.1 Create booking	Pending booking entry, New booking alert	D3 Booking database, Service Provider
Service Provider updates service availability	Service Provider	Services timeslot	3.4 Process availability	Updated services available time	D3 Booking database
Service Provider confirms or rejects booking	Service Provider	Booking confirmation response	3.2 Process provider response	Updated booking status	D3 Booking database
System notifies client of booking status	D3 Booking database	Booking Status record	3.3 Update booking status	Booking update message	Client

Figure 7.4.3 ERT of Process 3

Event	Source	Trigger	Activity	Response	Destination
Initiate Payment	Client	Click "Pay Now" button	Retrieve booking, compute total cost, and format transaction payload	Transaction_Payload	Process 4.2
Request Auth	Process 4.2	Bank API Request	Transmit payment parameters to bank portal	Payment_Request	Bank Portal
Verify Auth	Bank Portal	Bank Auth Response	Evaluate bank authentication response	Verified_Transaction_Data	Process 4.4
Finalize Transaction	Process 4.4	Payment Approved	Create payment record and update booking status to "Confirmed"	Status_Update	D3 Bookings / D4 Payments
Issue Receipt	Process 4.4	Database Success	Format receipt with transaction details	Digital_Receipt	Client / Service Provider

Figure 7.4.4 ERT of Process 4

Event	Source	Trigger	Process	Response	Destination
Client requests for reschedule	Client	Reschedule Request	5.1, 5.2, 5.3	React Frontend sends new date by JSON, Node.js Backend checks slot availability	eGHL SDK Module
Client requests for cancellation	Client	Cancel Request	5.1, 5.2, 5.3	React Frontend sends cancel command, Node.js retrieve booking data	eGHL SDK Module
Processing transaction refund	eGHL SDK	Fee amount variable passed	5.4	Executes HTTPS Redirect to handle refund	eGHL API Gateway / Client UI
Processing confirmed schedule change	eGHL Webhook	HTTP 200 OK (Refund)	5.5	Executes SQL UPDATE or DELETE command to alter slots in database.	D3 MySQL Database & Client UI

Figure 7.4.5 ERT of Process 5

Event	Source	Trigger	Activity	Response	Destination
Service Provider uploads bulk services data	Service Provider	Bulk service data file	Validates bulk services data, store inside Service Catalog and return upload status	Upload status notification	Service Provider
Service Provider configures custom buffer settings	Service Provider	Custom buffer settings	Fetches services list and updates custom buffer settings and if prompted updates the constraints for existing bookings	Configuration confirmation and AI-generated impact preview	Service Provider
Service Provider requests analytics	Service Provider	Analytics fetch request	Fetches and analyzes all services offered, booking performance data, and revenues data, generates dashboard view	Analytics dashboard view	Service Provider
Service Provider requests analytics with filters (date ranges/ months, services)	Service Provider	Analytics fetch request and filters	Fetches and analyzes all services offered, booking performance data, and revenues data, generates dashboard view	Filtered analytics dashboard view	Service Provider

Service Provider exports analytics	Service Provider	Clicks on “Export” button on analytics dashboard	Prepare analytics in PDF or CSV format, downloads report in user’s device	PDF/CSV file	Service Provider

Figure 7.4.6 ERT of Process 6

7.5 Structure Chart

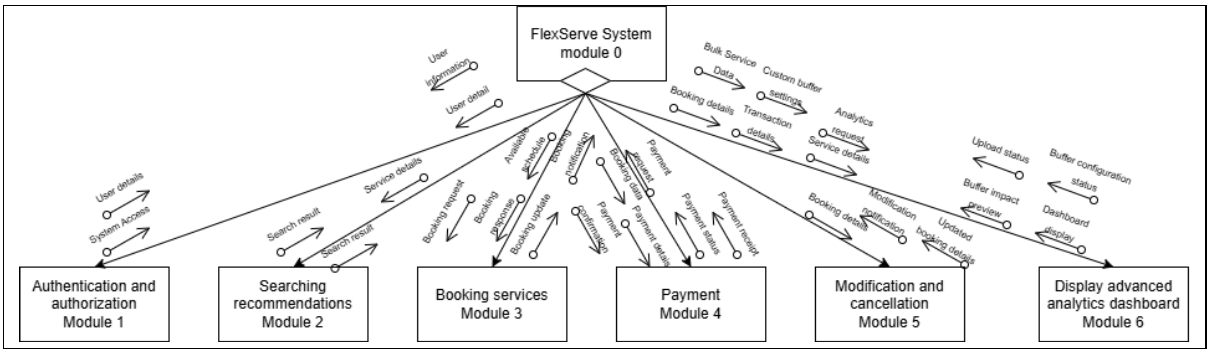

Figure 7.5.1 Structure chart of Diagram 0

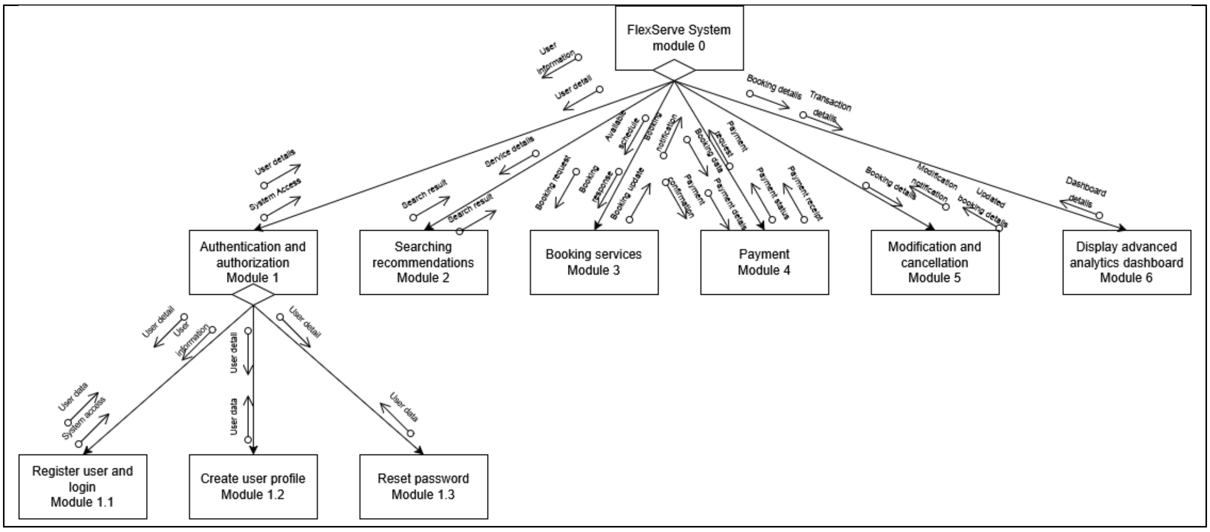

Figure 7.5.2 Structure chart of Process 1

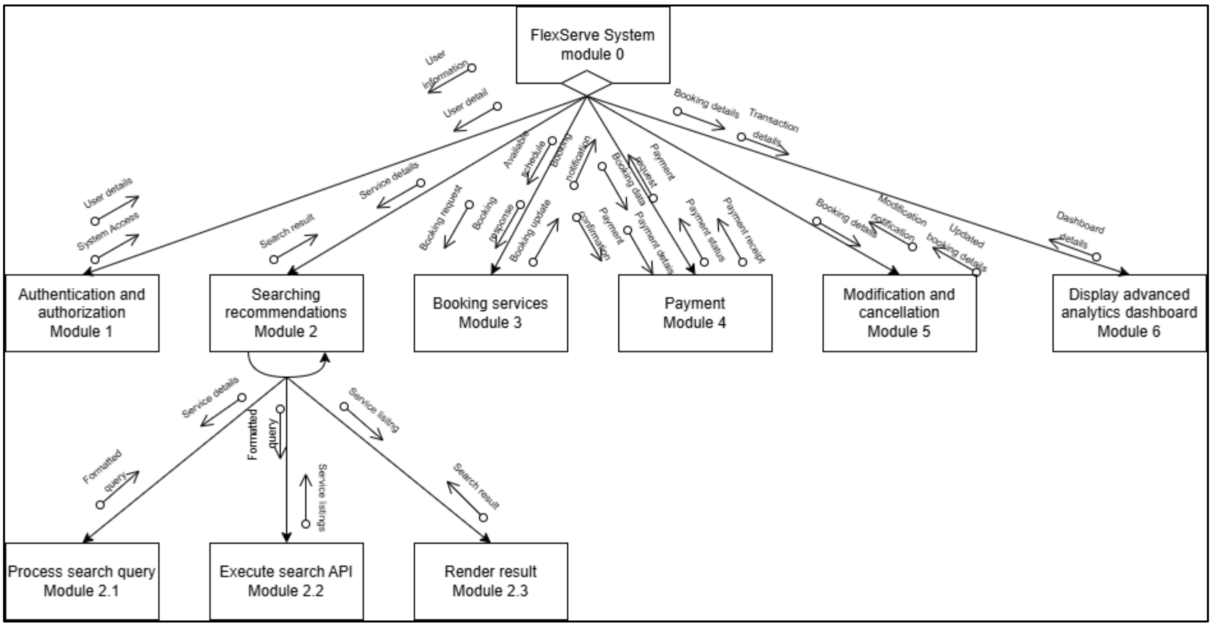

Figure 7.5.3 Structure chart of Process 2

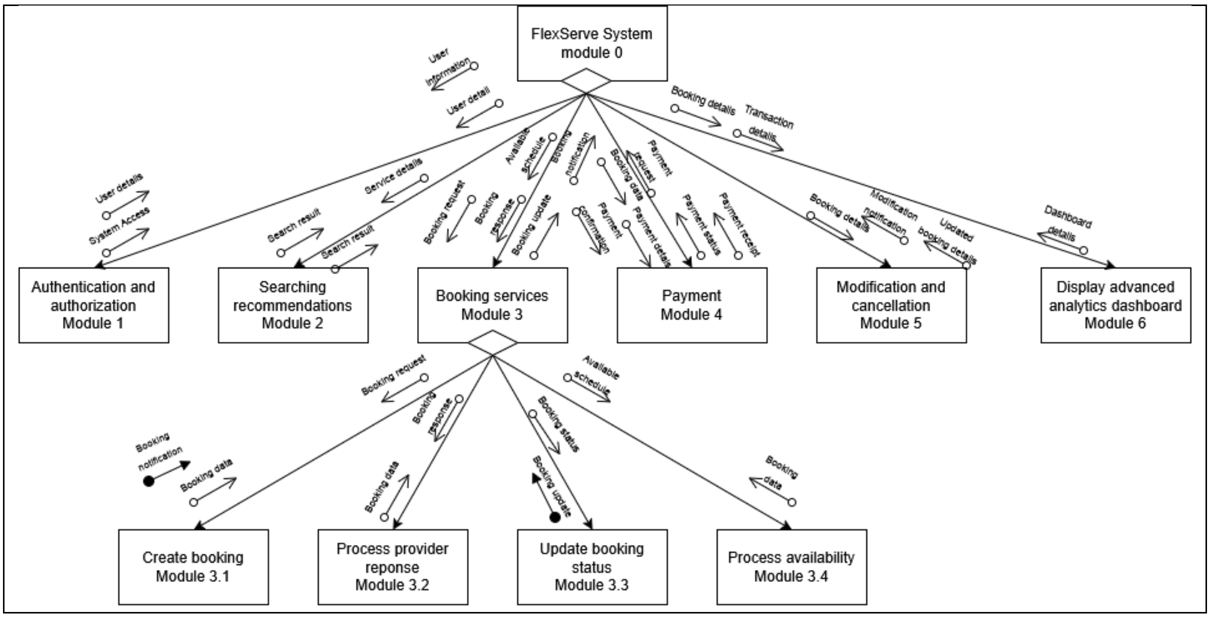

Figure 7.5.4 Structure chart of Process 3

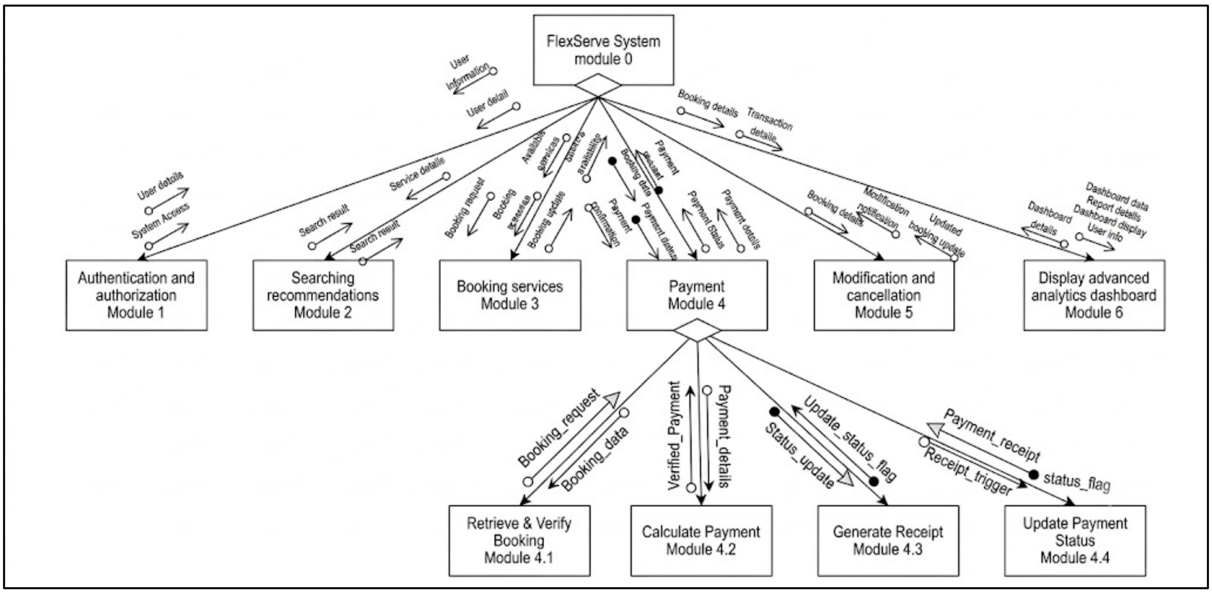

Figure 7.5.5 Structure chart of Process 4

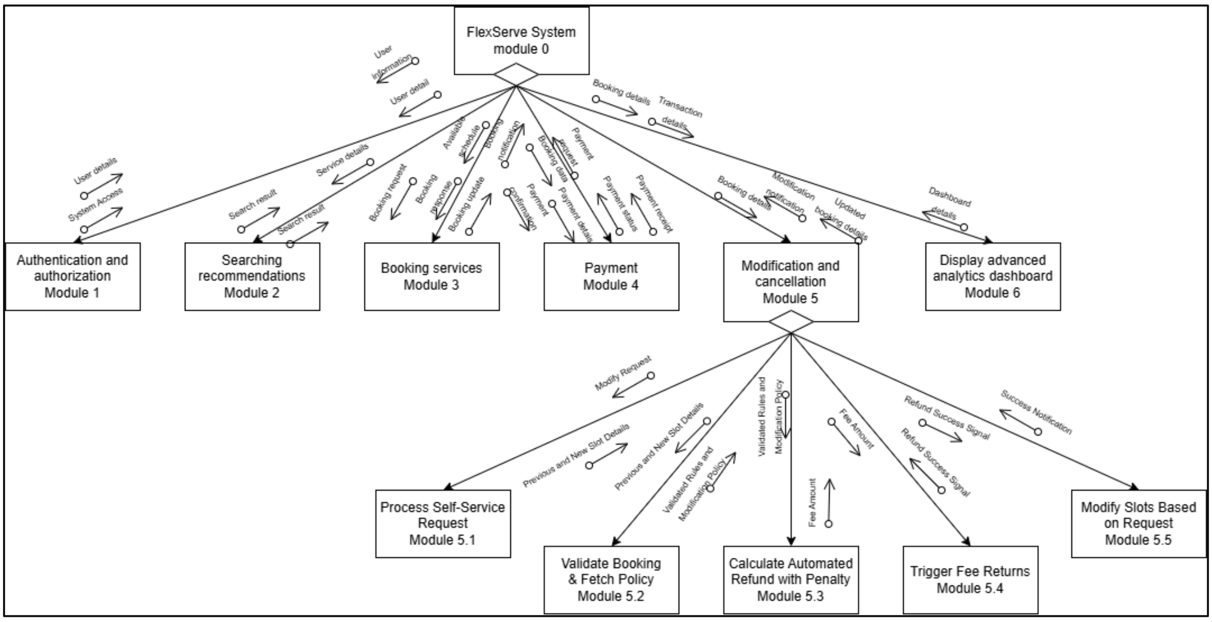

Figure 7.5.6 Structure chart of Process 5

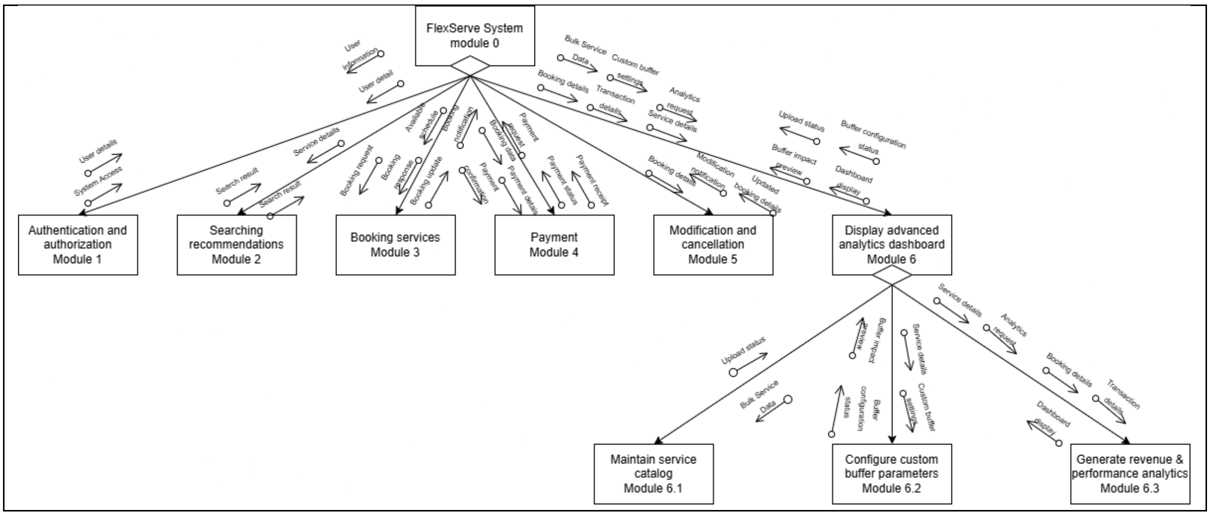

Figure 7.5.7 Structure chart of Process 6

7.6 System Architecture

Some of the functional requirements for this project can be carried out as below.

For Client Functional Requirement:

Functional requirement	Description
Registration and login	Users are able to create an account and log into the system.
Advanced Service Search	Users are able to search for services, filter based on specific needs, compare service details, prices, and reviews.
Direct Real-Time Booking	Users are able to view real-time availability and directly book open slots through the platform without waiting for manual confirmation.
Integrated Digital Payment	Users are able to pay booking or partial fees directly within the application during the checkout process.
Automated Receipt Retrieval	The system must generate and provide immediate digital/PDF receipts to the client upon successful transaction.
Self-Service Booking Modification	Users are able to request changes to a confirmed booking directly through the system.

The inputs, processes and outputs are stated below:

FR	Input	Process	Output
Registration and login	Username, password and email	Store information	Created account for user
Advanced Service Search	Search keywords, category filters, or minimum ratings	Match and filter user queries against the registered service database	Displayed list of matching services with prices and reviews
Direct Real-Time Booking	Selected service, desired date and time slot, user account details	Verify real-time slot availability in the system database	Reserved booking slot pending checkout
Integrated Digital Payment	Payment credentials, billing details, booking/partial fee amount	Securely process the financial transaction via the payment gateway	Transaction status marked as "Successful" or "Verified"
Automated Receipt Retrieval	Verified transaction data, client booking details	Generate an official digital invoice and compile data into a PDF format	digital/PDF receipt delivered to the client
Self-Service Booking Modification	Current Booking ID, requested new date/time slot	Validate availability of the new requested slots and update the schedule database	Updated booking schedule status and real-time confirmation update

For Provider Functional Requirements:

Functional requirement	Description
Registration and login	Users are able to create an account and log into the system.
Automated Booking Notifications	The system must instantly notify users when a new booking is made.
Dashboard Overview	Users are able to review booking details and verified payment statuses.
Provider-Initiated Cancellations & Adjustments	Users are able to cancel bookings or initiate schedule adjustments if unexpected issues arise.
Automated Monthly Earnings Reporting	The system must compile and generate automated monthly earnings reports for users.

The inputs, processes and outputs are stated below:

FR	Input	Process	Output
Registration and login	Username, password and email	Store information	Created account for user
Automated Booking Notifications	Completed and paid booking record details	Capture new booking events and instantly trigger system notification alerts	In-app notification or system alert delivered to the user
Dashboard Overview	User ID, booking history records, and payment gateway logs	Retrieve, compile, and structure user-specific booking and verified transaction data	Dashboard containing booking summaries and payment statuses
Provider-Initiated Cancellations & Adjustments	Booking ID, cancellation reason, or requested new schedule dates	Process the schedule cancellation or adjustment, update the database, and trigger automated refunds if necessary	Updated booking status ("Canceled" or "Rescheduled") and modification notifications
Automated Monthly Earnings Reporting	User transaction logs and aggregate clearance data for the current 30-day window	Calculate total earnings, aggregate processing fees, and format the data into a monthly report layout	Downloadable monthly earnings report document

For Core System Requirements:

Functional requirement	Description
Automated Availability Validation	The system must automatically audit the real-time availability of the time slots to prevent scheduling conflicts.
Integrated In-App Communication	The system must provide a built-in communication channel.
Automated Status Updates & Syncing	The system must instantly sync the schedule for both the client and provider accounts when a booking is modified or canceled.
Automated Cancellations & Refunds	The system must automate the cancellation from processing refunds back to the customer's side when a booking is aborted.
Interactive Buffer-Time Configuration	The system must inform providers to set up preferred "Buffer Times" (e.g., cleanup or travel time). It should also analyze scheduling patterns over time to autonomously optimize and suggest buffers over time.

The inputs, processes and outputs are stated below:

FR	Input	Process	Output
Automated Availability Validation	Requested appointment date, time slot, and provider calendar database	Query the database to audit and verify if the requested slot is free	Validation status
Integrated In-App Communication	Sender ID, receiver ID, and typed text message or attachment data	Process, route, and deliver the message packets to the designated recipient through a secure, built-in channel	Real-time chat dialogue
Automated Status Updates & Syncing	Modified or canceled booking records and associated User IDs	Update the master database and synchronize schedule data	Refreshed calendar views and automated notifications
Automated Cancellations & Refunds	Booking record details and original payment transaction tokens	Process the system cancellation routine, calculate appropriate refund rules, and trigger the payment gateway	Canceled booking status update and refund transaction confirmation
Interactive Buffer-Time Configuration	Manual buffers and historical booking pattern timelines	Analyze recurring booking habits to calculate and suggest ideal time limits	Buffer locked into the active calendar schedule

Non-functional Requirement

Some of the non-functional requirements for the project can be carried out as below:

Non-functional requirement	Description
Performance	The system is able to load and display the details quickly after a search.
Availability	The system is able to operate well throughout the whole time.
Security	The system is able to store and secure both users and service providers credentials.
Accuracy	The system is able to show real-time information to the users about service providers details.
Compatibility	The system is able to operate regularly on common devices.

8. System Wireframe

Figure 8.1.1 Login Page

Figure 8.1.2 Unsuccessful Login

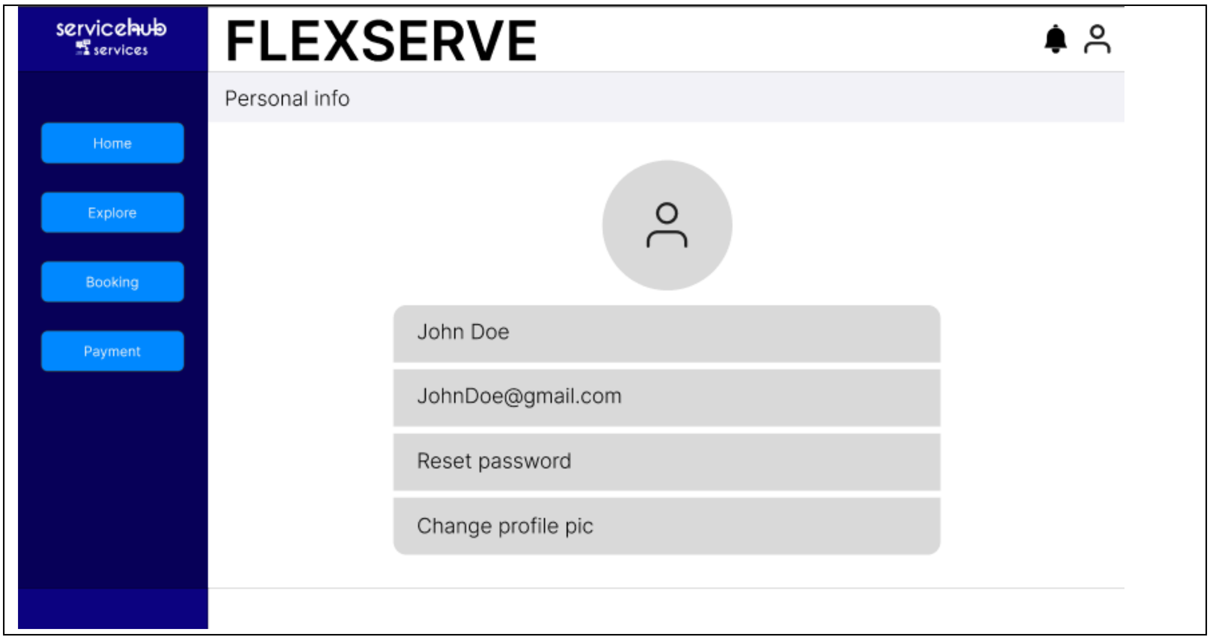

Figure 8.1.3 Setup user profile

Figure 8.1.4 Reset password

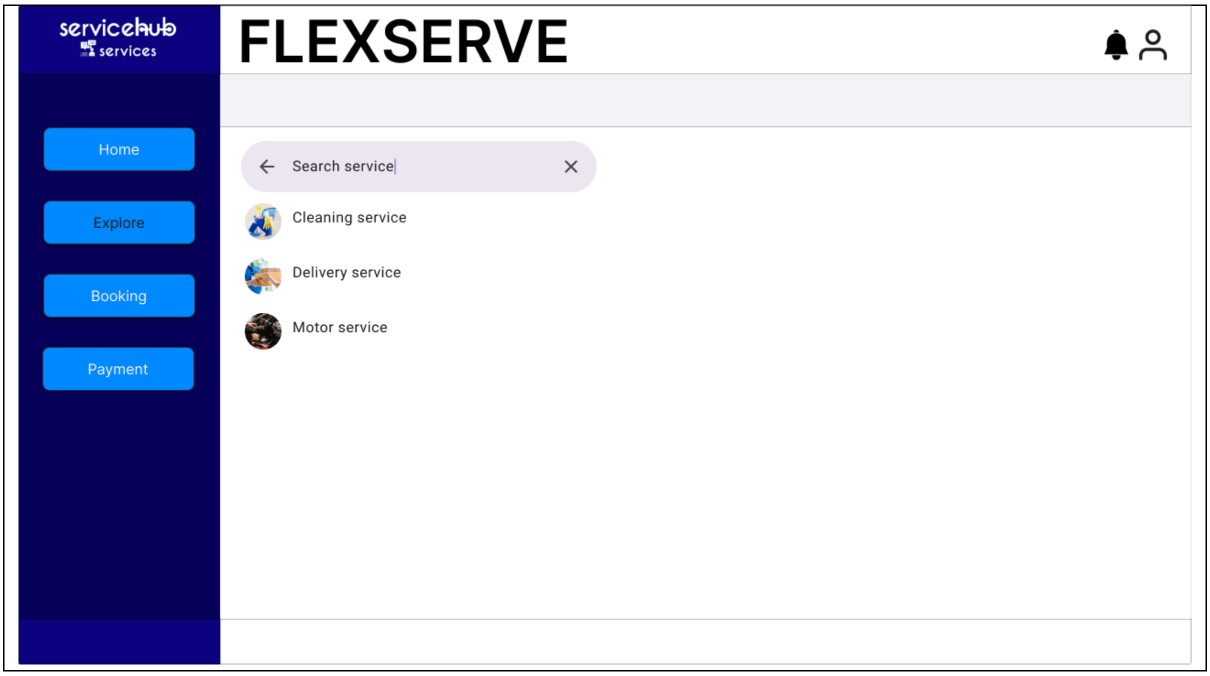

Figure 8.2.1 Searching

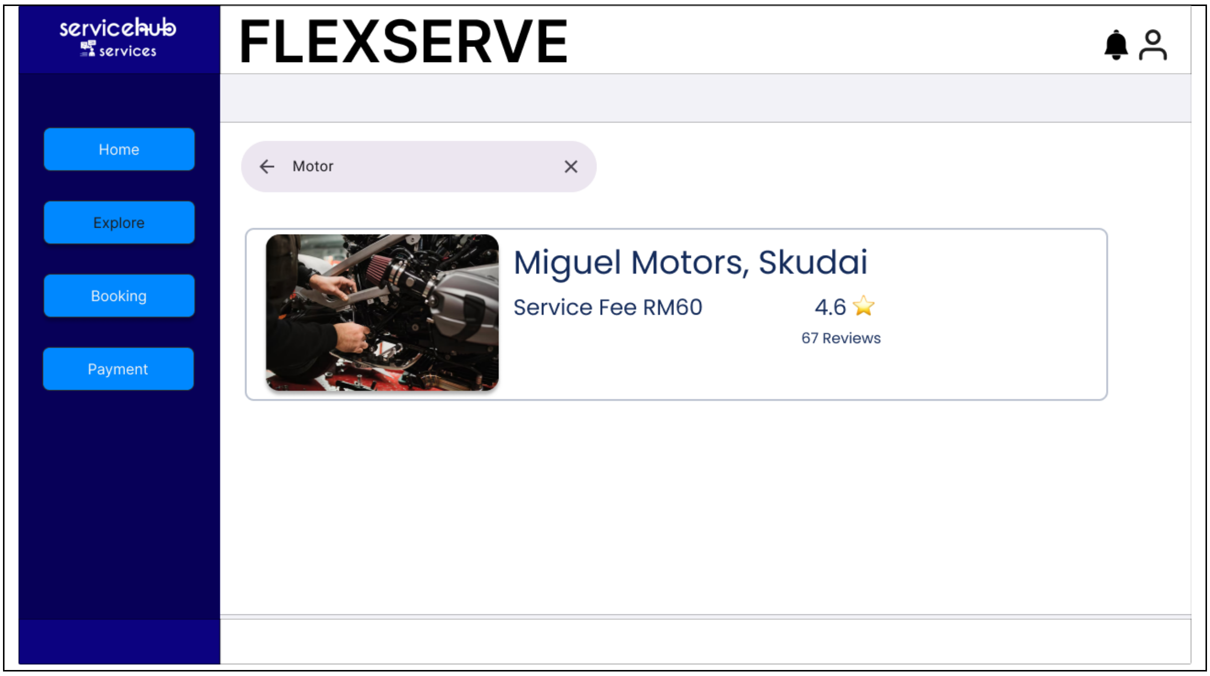

Figure 8.2.2 Display Result

Figure 8.2.3 No Result Found

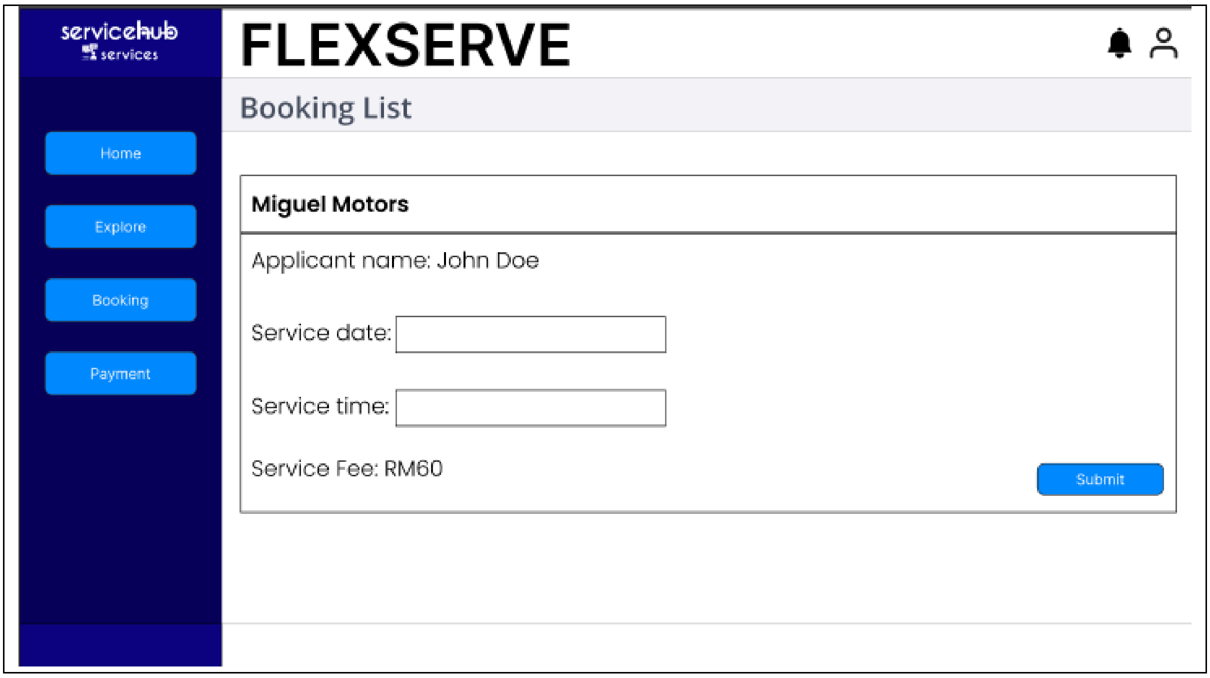

Figure 8.3.1 Client create booking

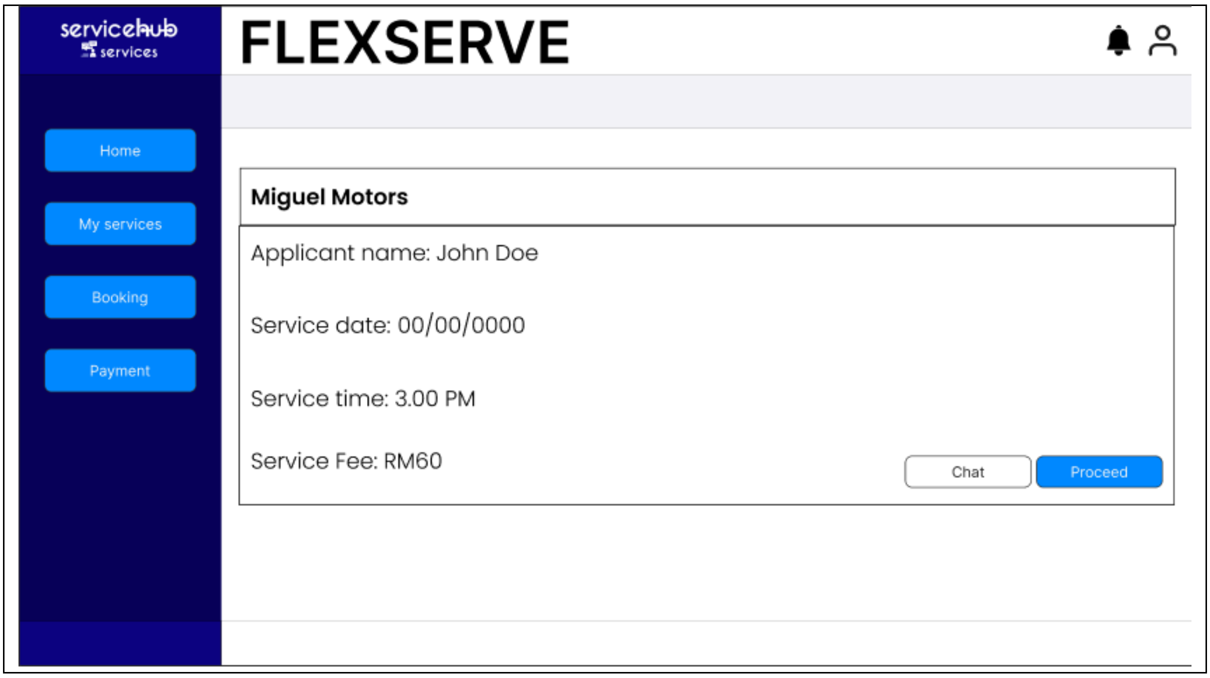

Figure 8.3.2 Service provider receive booking request

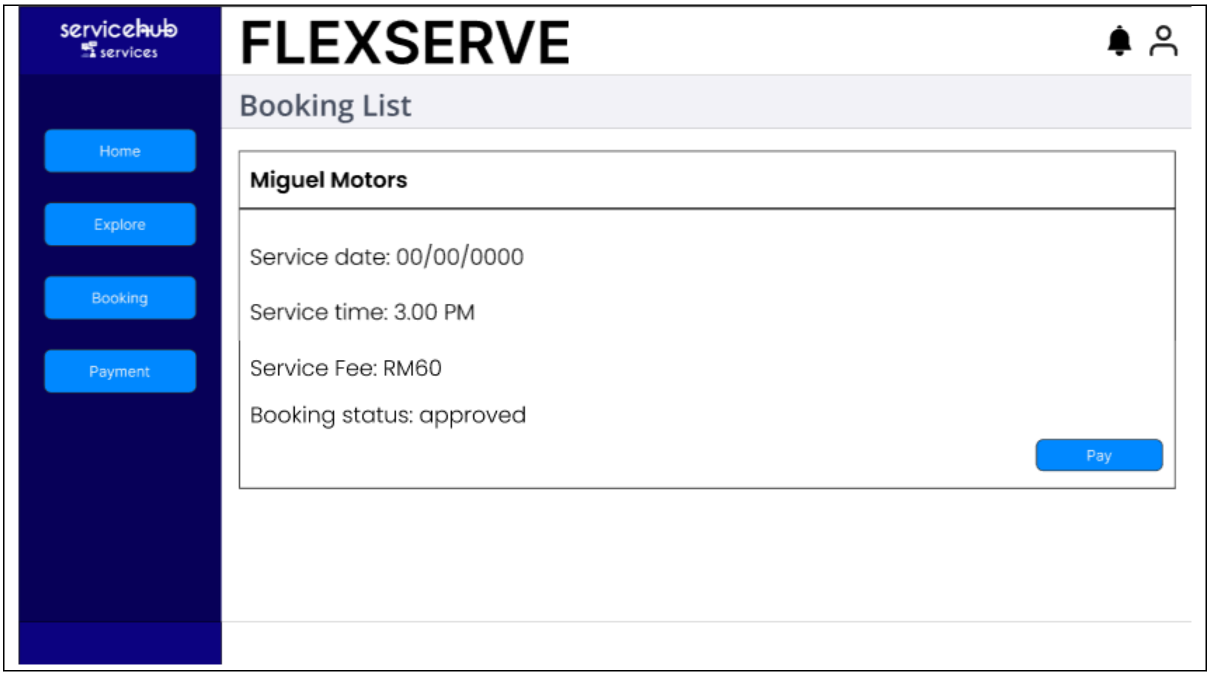

Figure 8.3.3 Client receive booking update

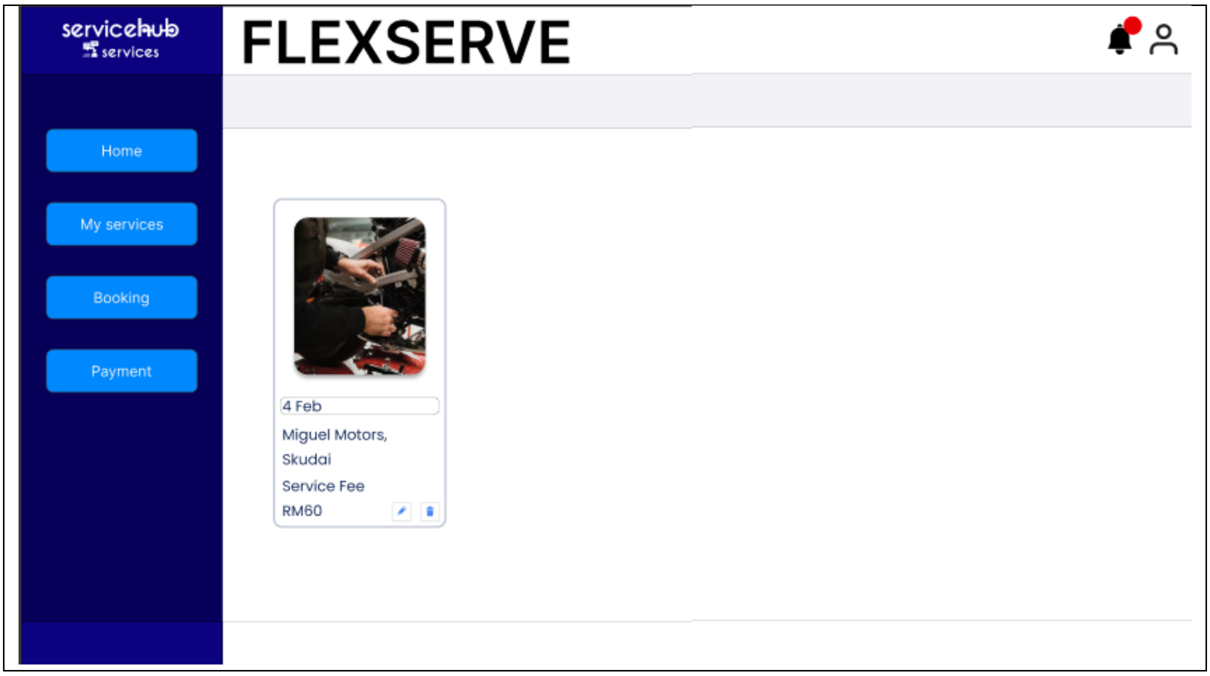

Figure 8.3.4 Service provider insert available slot time

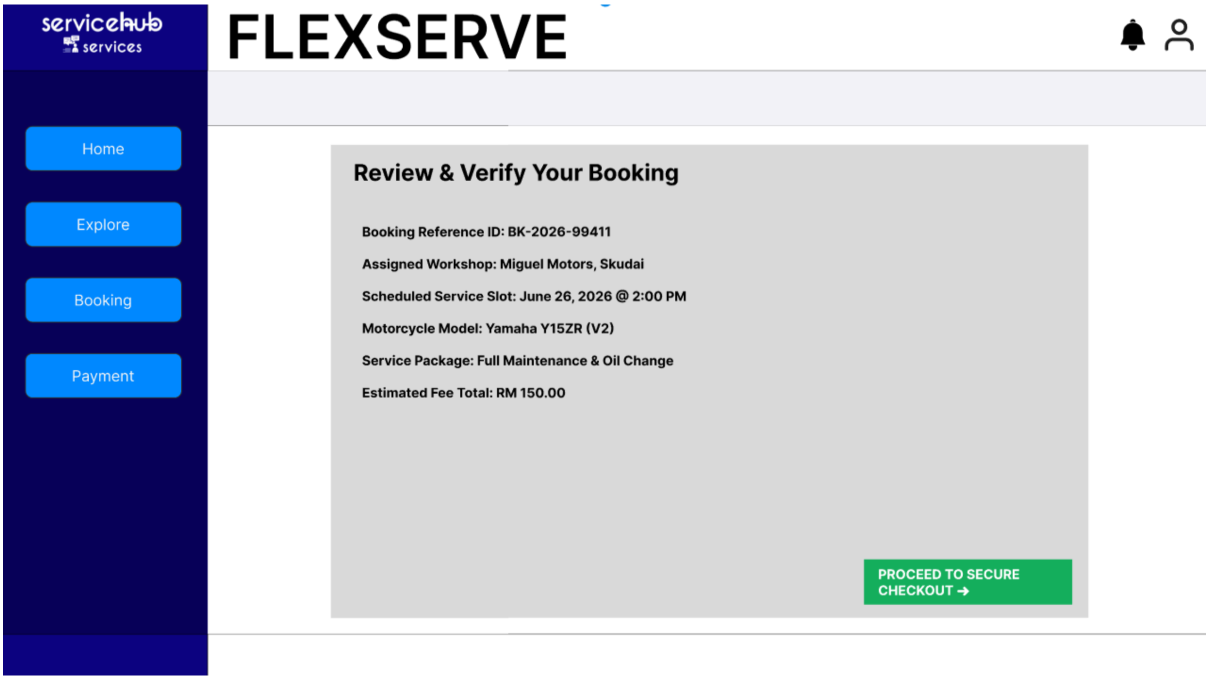

Figure 8.4.1 Verify Booking Details

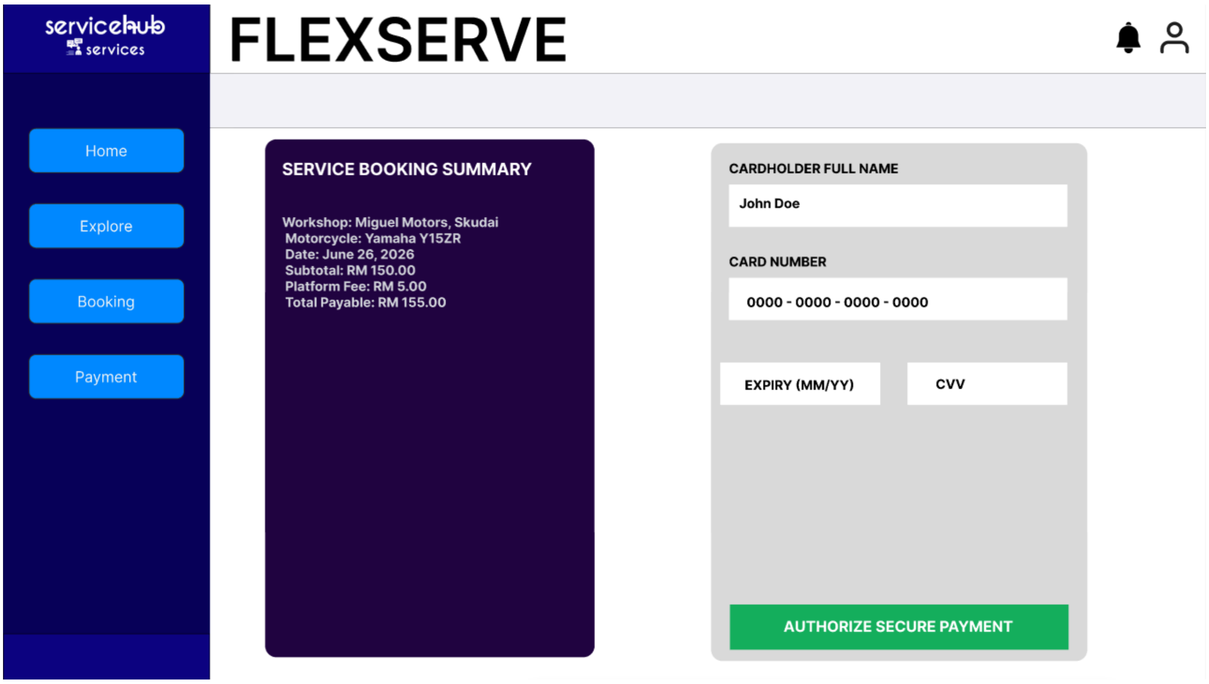

Figure 8.4.2 Secure Checkout Form

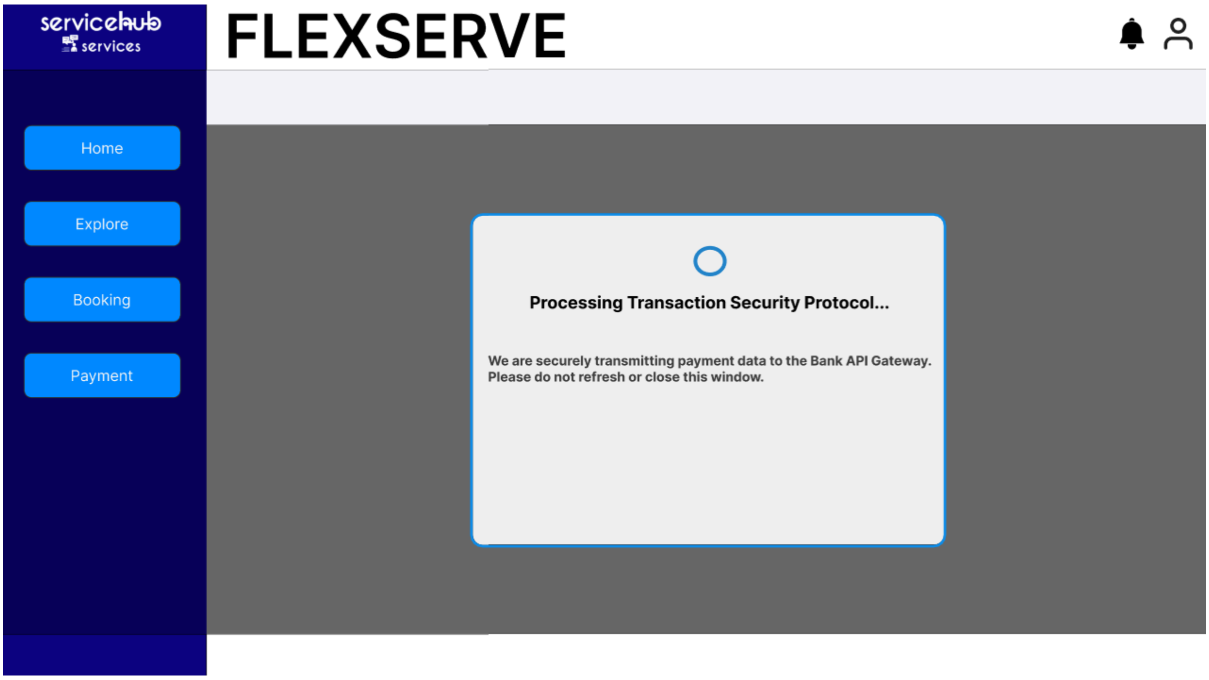

Figure 8.4.3 Bank API Gateway Processing

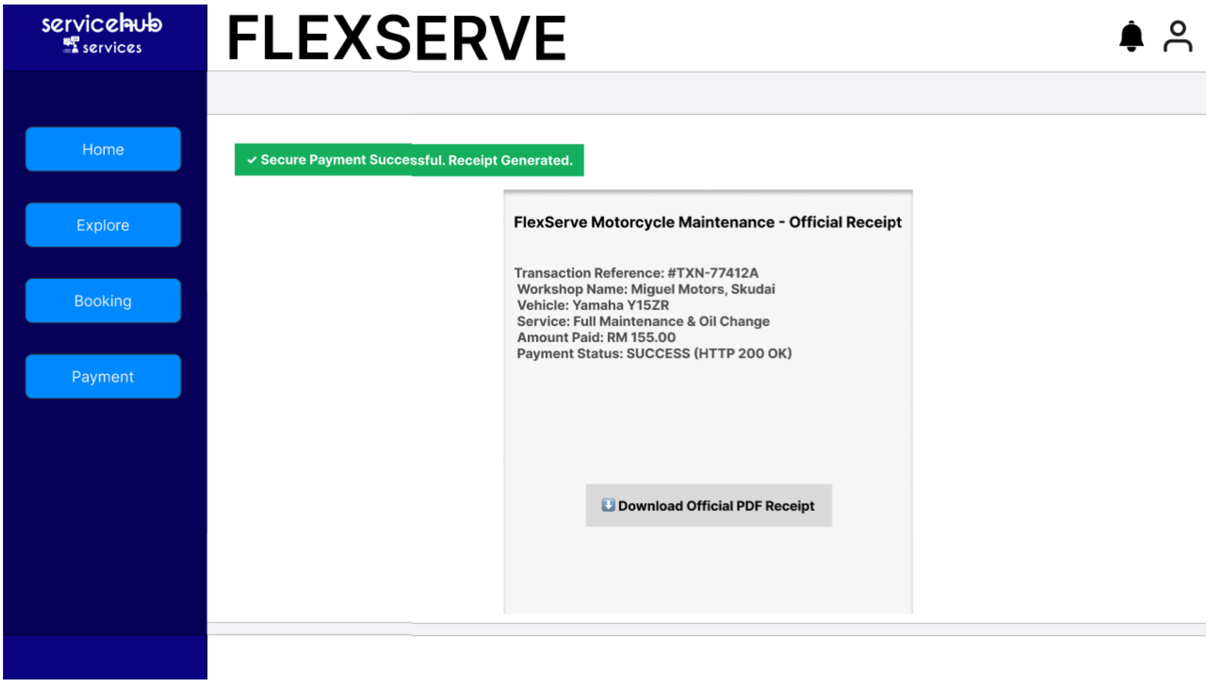

Figure 8.4.4 Transaction Success & Receipt

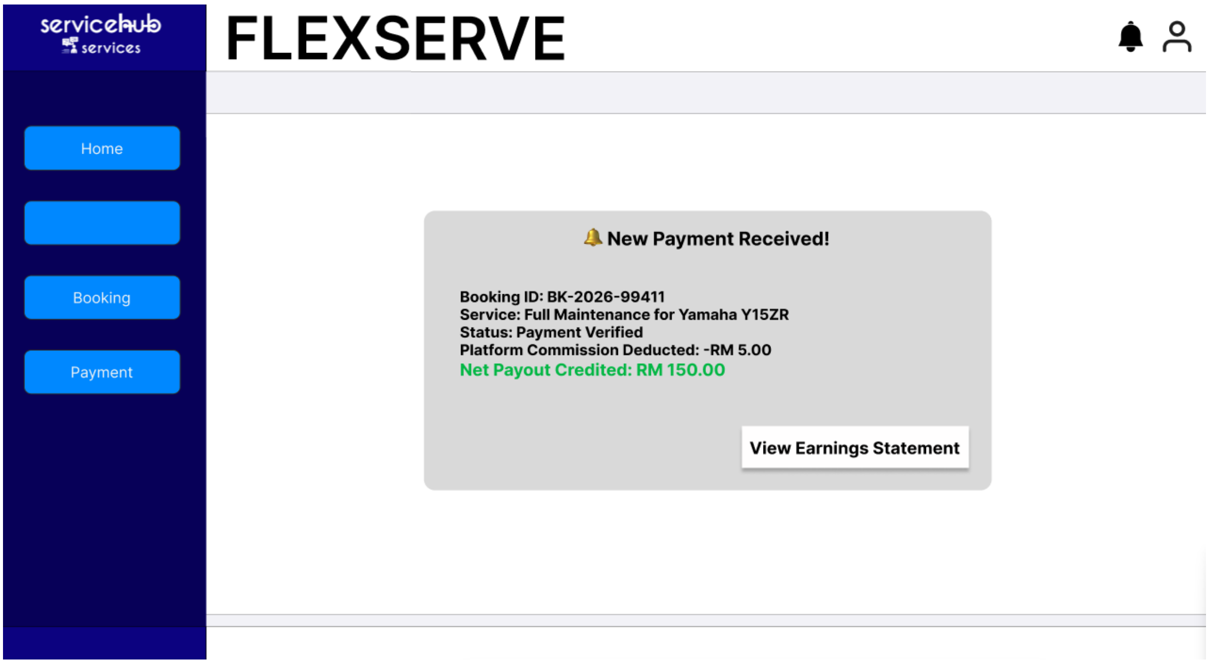

Figure 8.4.4 Provider Payment Notification

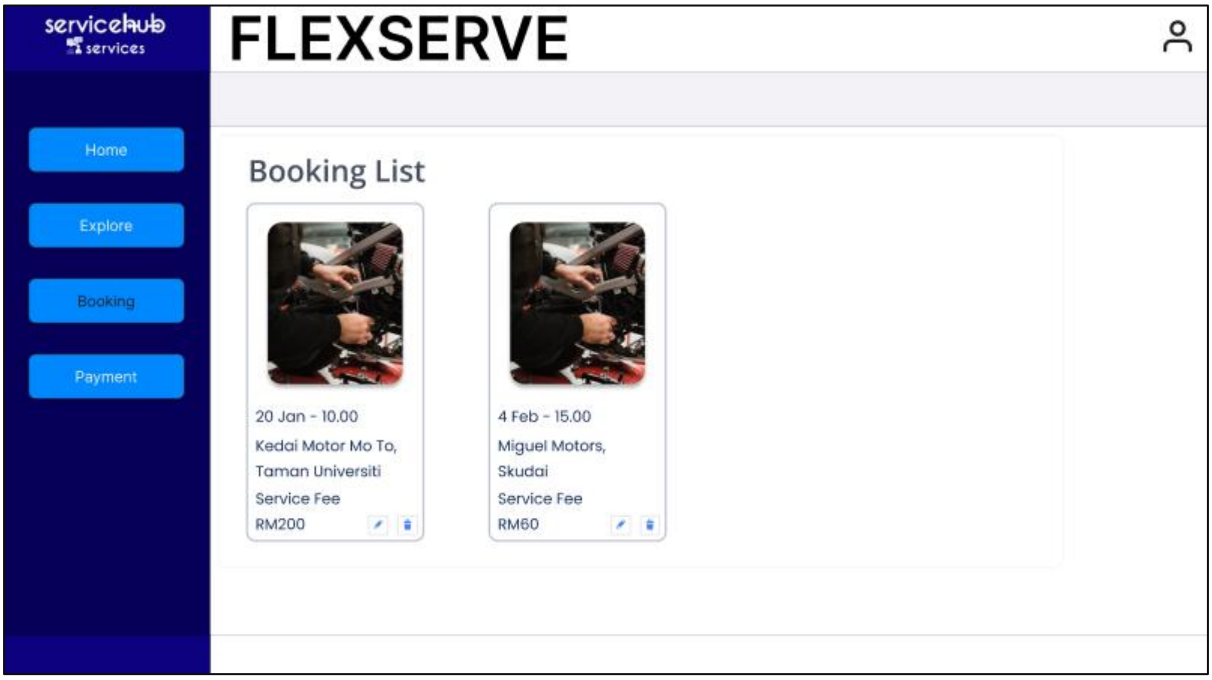

Figure 8.5.1 Booking dashboard

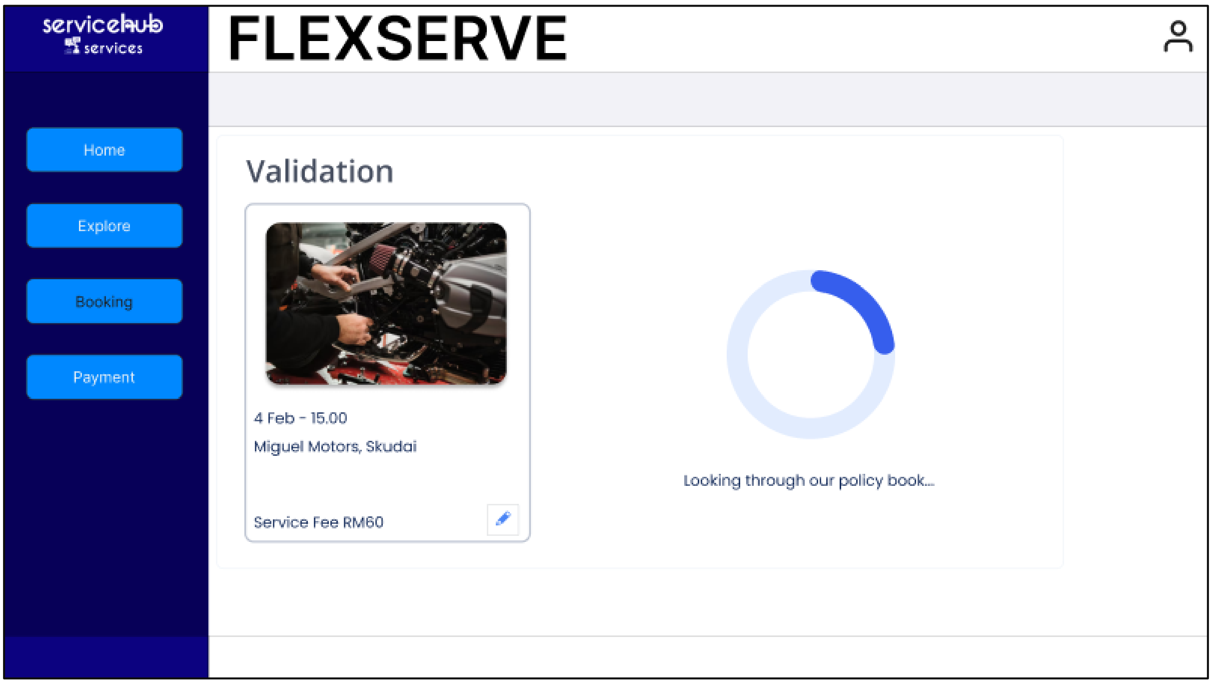

Figure 8.5.2 Modification validation

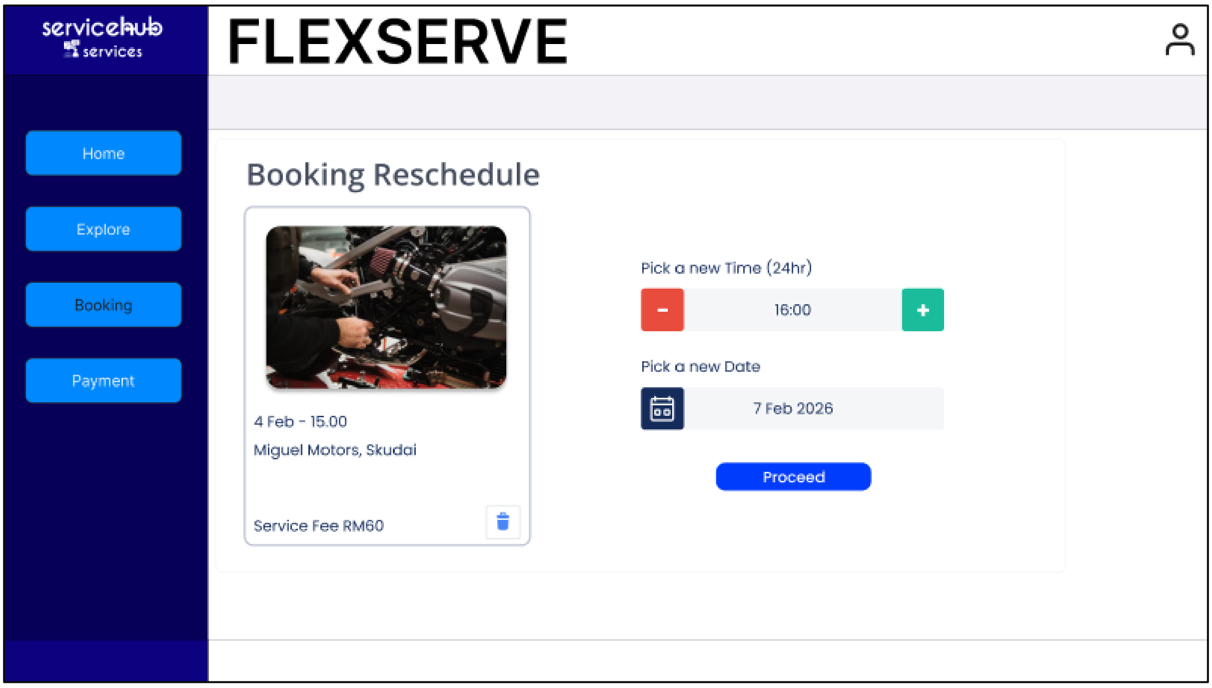

Figure 8.5.3 Reschedule

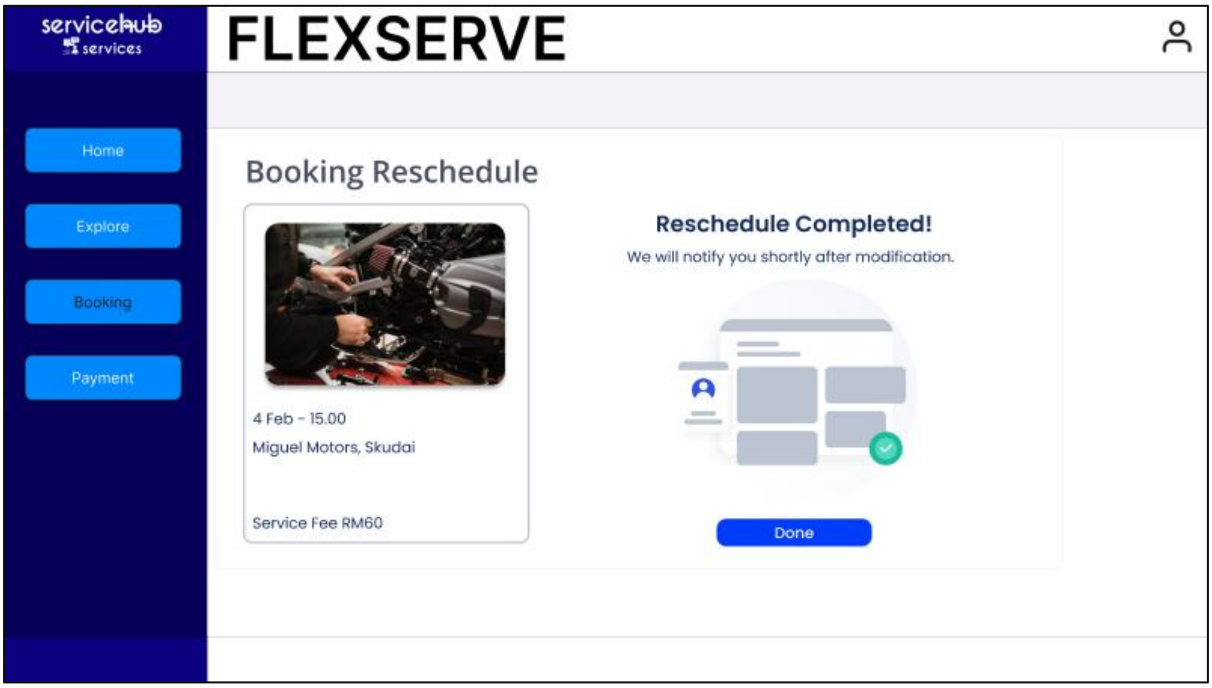

Figure 8.5.4 Reschedule successful

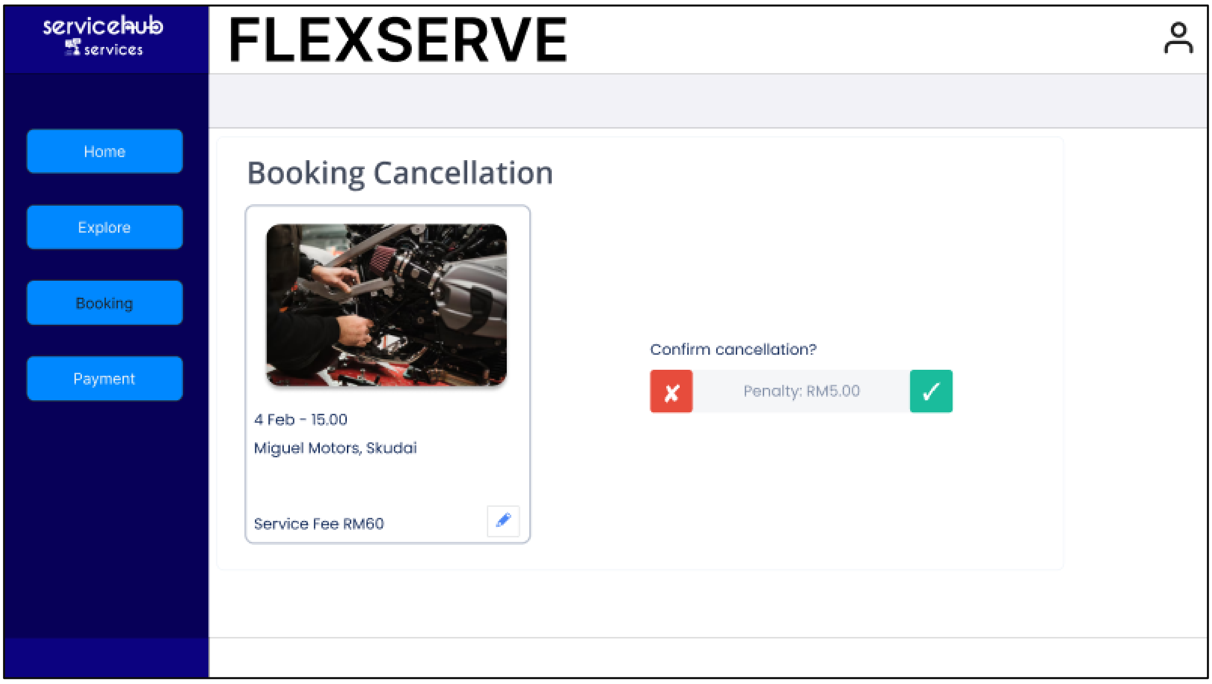

Figure 8.5.5 Cancellation confirmation

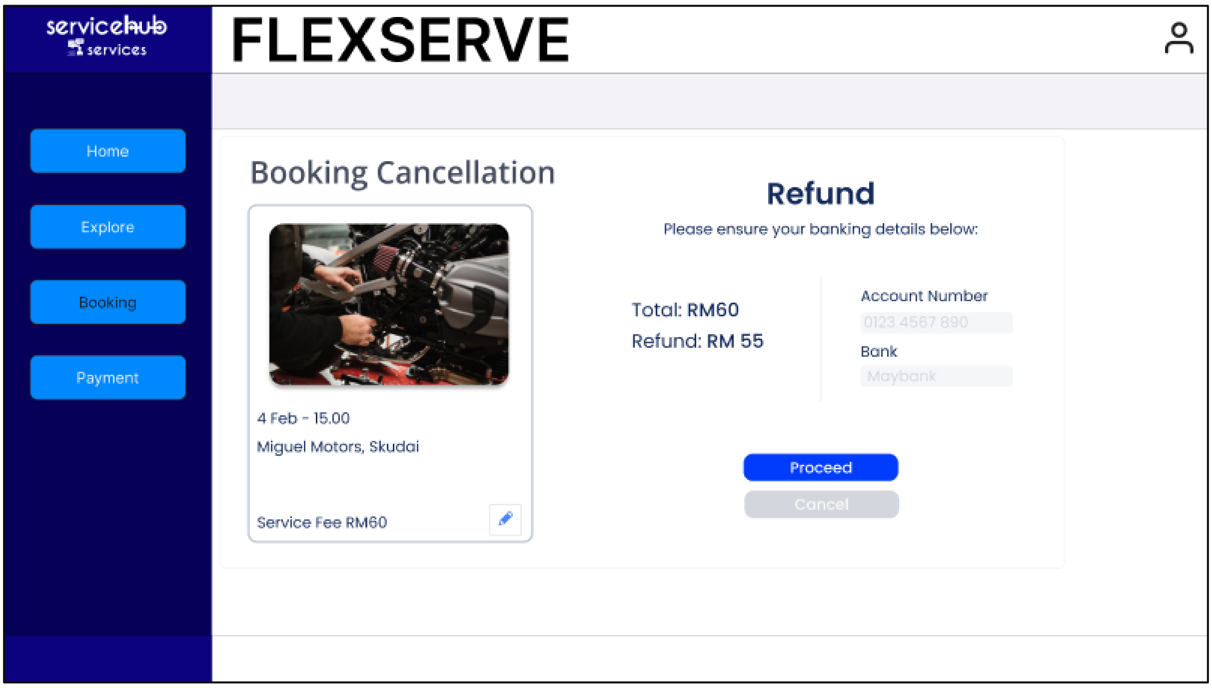

Figure 8.5.6 Refund process

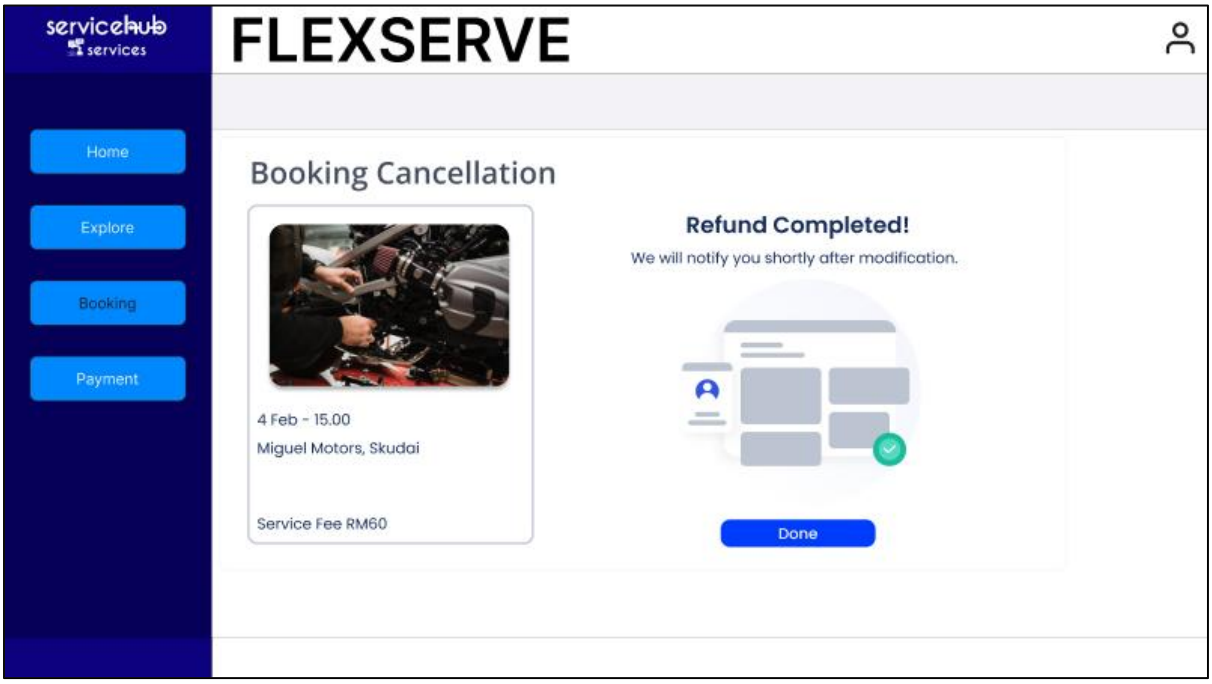

Figure 8.5.7 Refund success

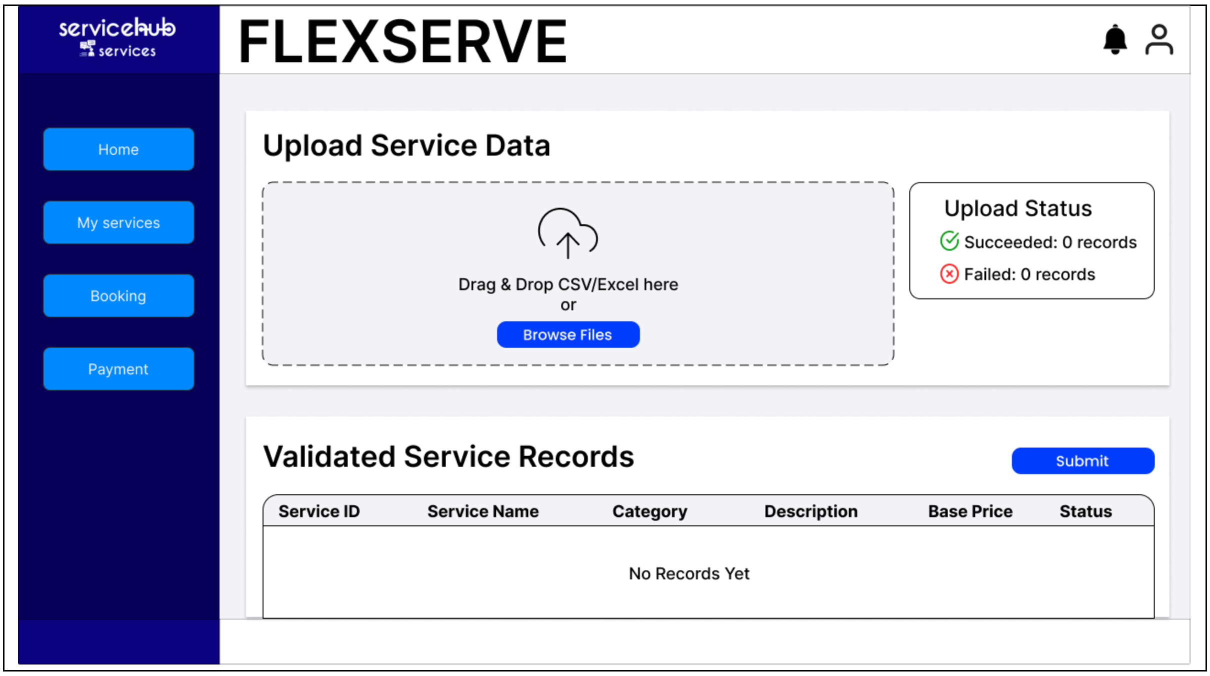

Figure 8.6.1 Process 6.1 Upload Services Page

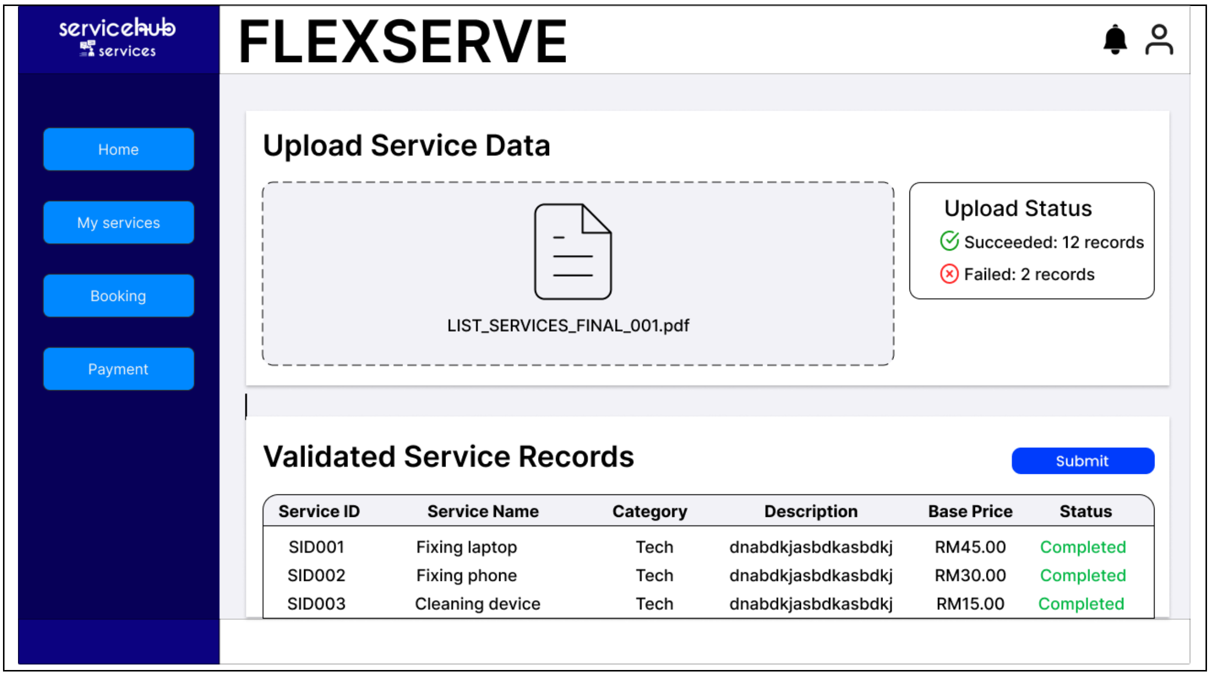

Figure 8.6.2 Process 6.1 Upload Services Page Output

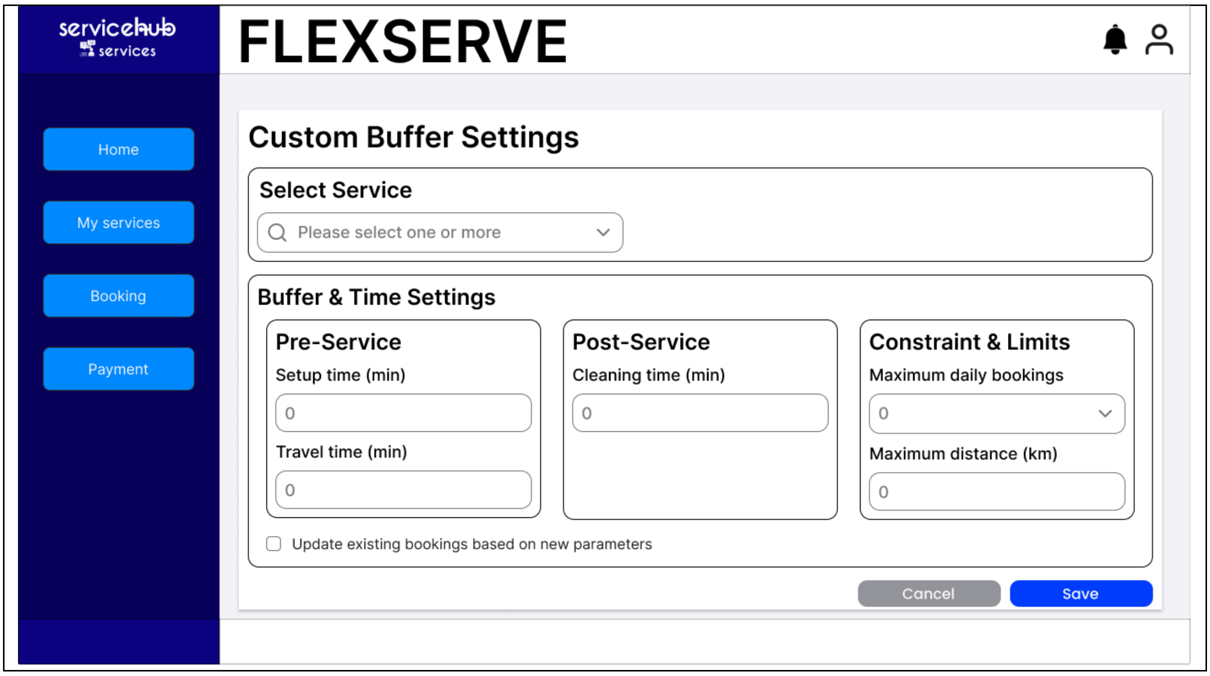

Figure 8.6.3 Process 6.2 Custom Buffer Form

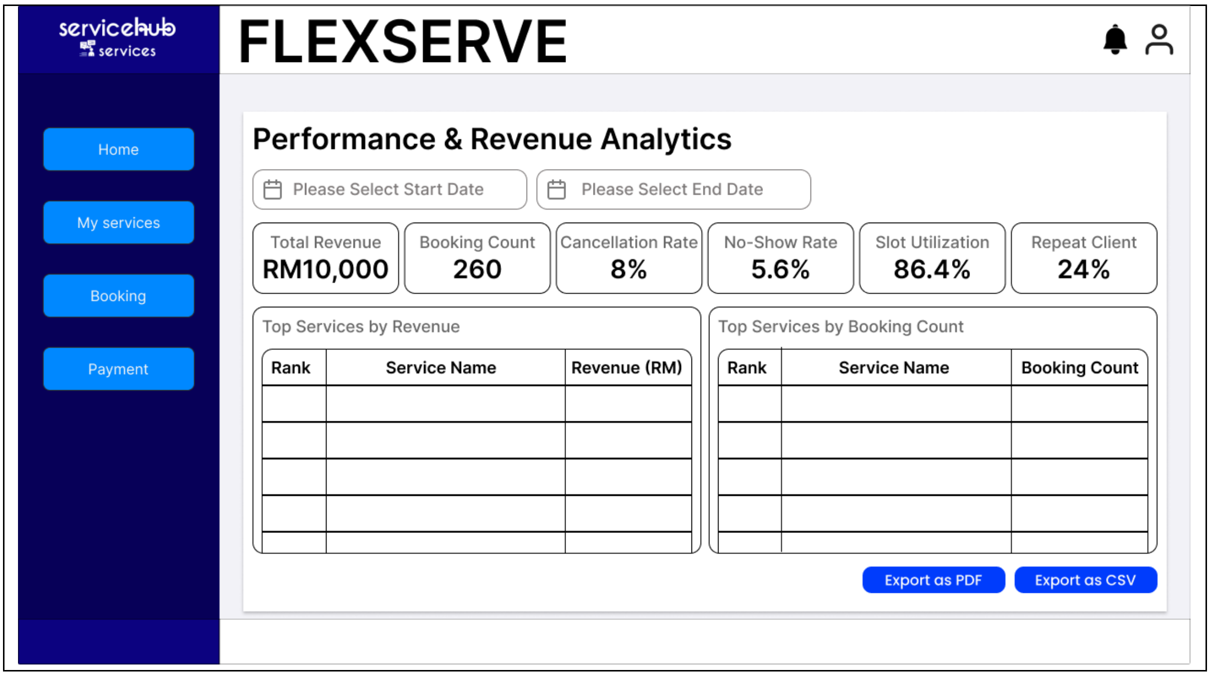

Figure 8.6.4 Process 6.3 Analytics Display

9. Summary of the proposed system

The development of FlexServe successfully solved the inefficiencies identified in the AS-IS analysis. The system eliminates reliance on fragmented processes such as manual booking using messaging apps like WhatsApp and manual scheduling that often leads to double booking problems.

The TO-BE system includes an AI Smart Scheduler that will directly solve double booking and scheduling conflict problems from the current system. Furthermore, the payment pipeline through the eGHL payment gateway replaces the manual bank transfer that was susceptible to delay and error. Clients won’t need to screenshot receipts and send them via chat as payment proof. The integrated payment gateway ensures a smooth transaction flow. Next, the self-service modification and cancelation module with automated refund removes the need for service providers to manually process refunds to the client’s bank account. Finally, the analytics dashboard with service upload and report provides an entirely new improvement that was not available in the current system. This dashboard will give service providers an insight into their revenue, performance, cancellation rates, and many more. This helps service providers with better decision making for their business.

Several suggestions for future improvements to the system are recommended. The future version should expand payment support to include multiple gateways beyond eGHL such as e-wallet options. This will improve accessibility for clients who prefer using e-wallets. Next, the implementation of a dedicated review system module to verify reviews and prevent fake ratings. This will further improve trust between client and service providers on the platform. Lastly, a mobile application would help increase FlexServe’s reach beyond just web browser users. Furthermore, all users will gain more access to the system, and many processes will be more efficient with mobile phones to push notifications to alert clients or service providers.